Industrial Electronics N5

Module 3

Operational Amplifiers

Learning outcomes

On completion of this module, you will be able to:

3.1 Transistor amplifiers

- Discuss circuit diagrams and operating principles of biasing and stabilization in a single stage common emitter amplifier
- Explain the term stabilization
- Draw neatly labelled circuit diagrams of fixed bias circuit, self-bias and voltage diver bias circuit
- Perform calculation of the voltages, currents, resistances and gain of the fixed bias circuits
- Depict the graphical representation and description of amplitude, non-linear, and frequency distortion
- Draw circuit diagrams of the transistor equivalent and h-equivalent circuits of the common base and emitter amplifiers
- Calculate:
    - Voltage, current power gain and impedance
    - Component values for a single stage common emitter amplifier

3.2 Operational amplifiers

- Describe the characteristics of an operational amplifier.
- Draw the IEC circuit symbol of an operational amplifier.
- Draw the pin-layout for a 741 dual-in-line (DIL) 8-pin operational amplifier.
- Draw the pin-layout for a 741 flat-pack 14-pin operational amplifier.
- Draw the pin-layout for a 741 metal can 8-pin operational amplifier.
- Explain, with the aid of a circuit diagram, the operation of a basic differential amplifier.

3. TRANSISTOR AMPLIFIERS

3.1 INTRODUCTION

A transistor is a current amplifier, the small current at the base results in an amplified current at the collector. This Syllabus will be focused on the transistor from a DC and an AC point of view. The DC point of view concerns the characteristics and calculations of DC-amplifiers, while the AC concerns the small-signal analysis.

Watch the video below for more information on transistors: https://www.youtube.com/watch?v=l-rfPfRgNJ8

3.2 FIXED BIAS

FIGURE 3.1 displays the circuit diagram of a fixed bias common emitter amplifier. This circuit has only one supply voltage, which means the same bias is applied to both the base and the collector.

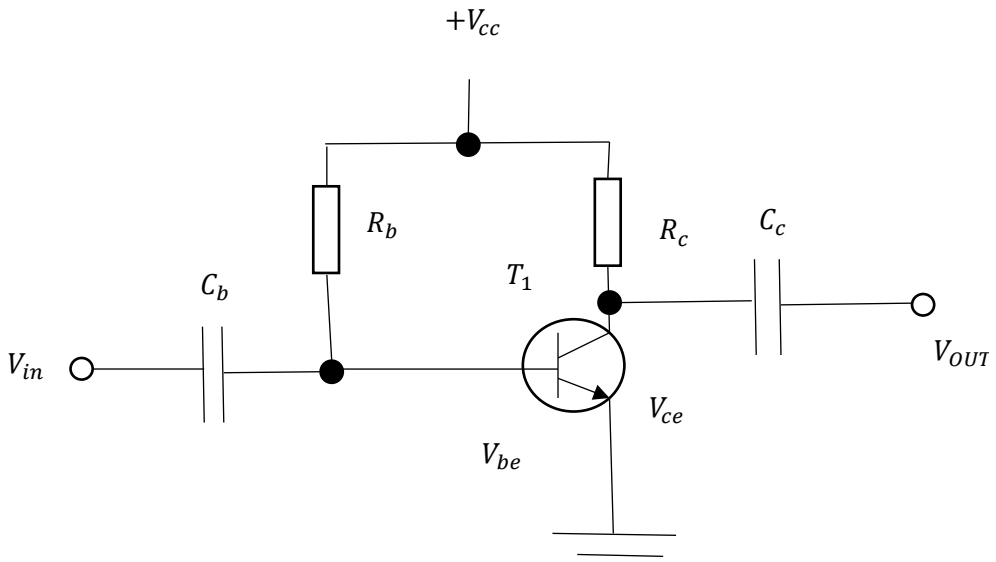


FIGURE 3.1 Fixed bias amplifier

BIASSING

- The transistor in FIGURE 3.1 can be used as an amplifier if the base-emitter junction is forwardly biased and the base-collector junction is backwardly biased. Thus V_{cc} must be positive to the ground

- As soon as the collector-emitter voltage V_{be} The base-collector junction will be forward biased, and the device will no longer function as an amplifier. The transistor will then be ***saturated***

- The base collector junction must be reversely biased thoroughly to ensure that the transistor works as a linear amplifier
- $\boldsymbol{R_b\ and\ R_c}$ Must thereby be calculated in such a way that the transistor never enters the saturated region, which means that V_{cc} must always remain greater than V_{be}. The bias of this circuit will therefore be dependent on the current gain or β -gain of the transistor. The relation between the collector and the base current $\boldsymbol{I_C = \beta I_b}$ is only valid as long as the transistor operates in the linear region

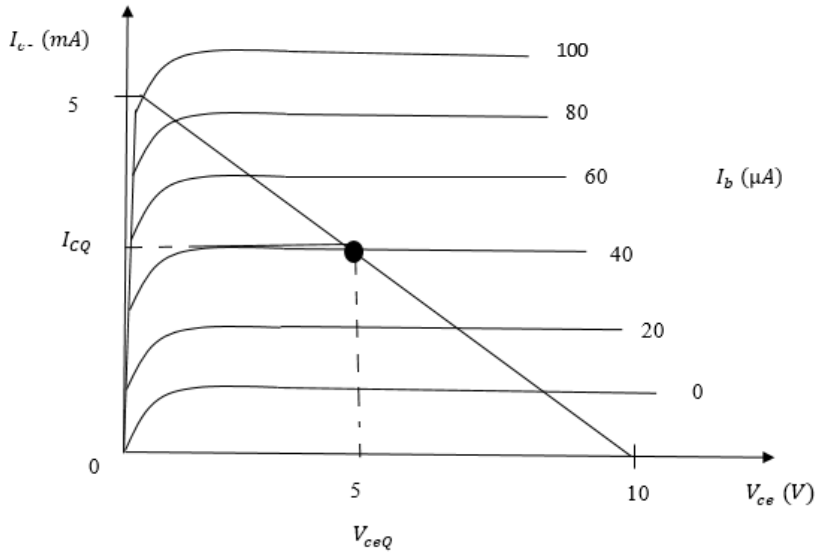
Watch the video below for more information on biasing:
https://www.youtube.com/watch?v=yLEcCtqsa14

3.3 STABILISATION OF FIXED BIAS

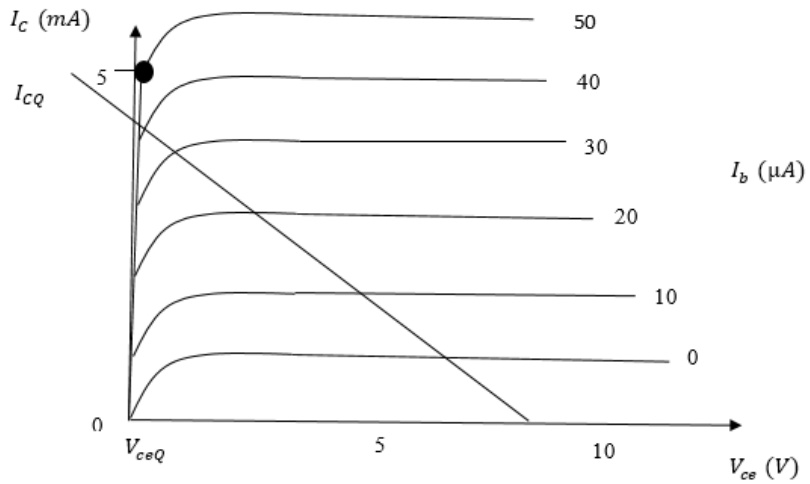
The fixed bias circuit produces a reasonable gain, as does the amplifier, but experiences problems with maintaining bias stability. In any amplifier, the collector current I_c will vary with change in temperature ***because of the following conditions***:

- ***Inverse saturating current*** (leakage current) doubles for each 10° increase in temperature
- ***Base emitter voltage***, V_{be} which decreases with 2,5milli Volts/C
- ***Current gain***, β, which increases with temperature

These factors create a shift in the work point because of the temperature change. This can be seen in FIGURE 3.2 A.) And B.)



A.) Work points at normal temperature ($25°C$)



B.) Work point at high temperature ($100°C$)

FIGURE 3.2

- ***To prevent this shift of the work-point***, an emitter resistor R_e is connected between the emitter and ground
- ***The function of R_e is to limit the collector's current I_c during an increase in temperature***
- The addition of R_e increases the bias stability, but oscillations originate in the circuit because this accomplishes negative feedback
- ***If the temperature should rise, I_c and β will increase***; a voltage drop V_e develops across R_e; this causes a decrease in β ; I_c decreases; V_e decreases; β decreases ; I_c increases ; V_e increases. Hence, the oscillations or AC – component
- This AC – component can be short-circuited by connecting an emitter capacitor C_e with R_e in parallel

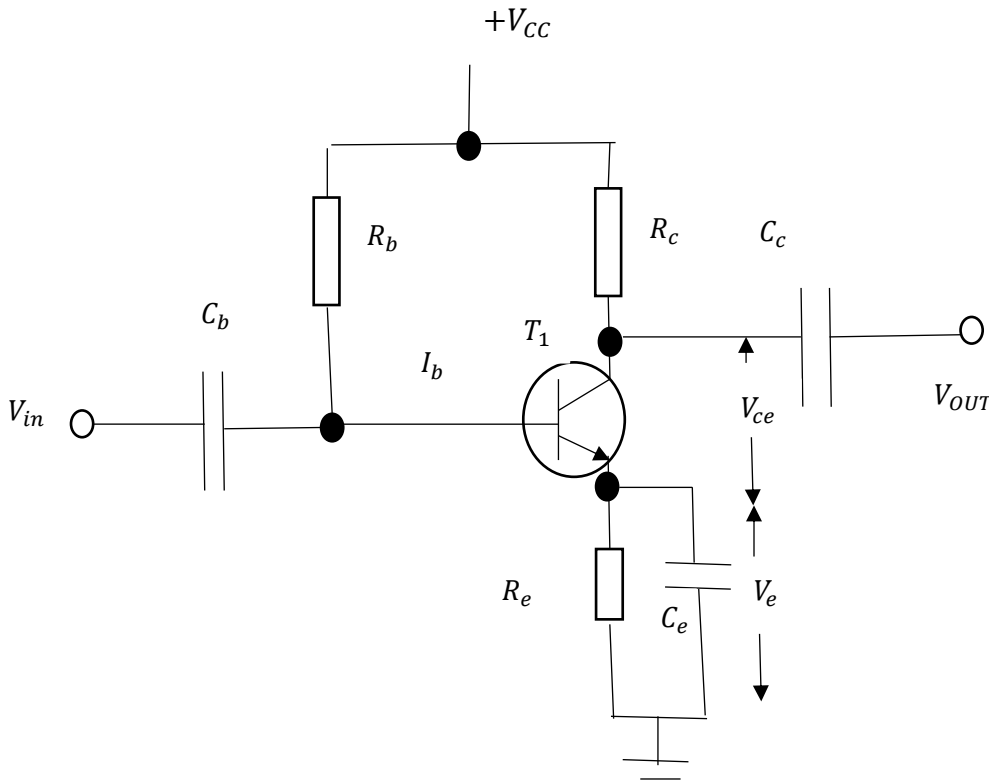


FIGURE 3.3 Fixed Bias amplifier with temperature stabilization

3.4 DESIGN OF A FIXED BIAS AMPLIFIER

The design of a fixed bias amplifier involves calculating the external components such as the collector resistor, the base resistor, and the capacitors to achieve amplification. For a fixed bias amplifier without a collector resistor, the following formulas are used for the resistors

$$R_c = \frac{V_{cc} - V_{ce}}{I_C} \; and \; R_b = \frac{V_{CC} - V_{be}}{I_b}$$

In this circuit, there is no emitter resistor R_cAnd the voltage V_{cc} is across the collector and ground

Example 3.1

Calculate the values R_c and R_b in FIGURE 3.4, (Note: A silicon transistor is used.)

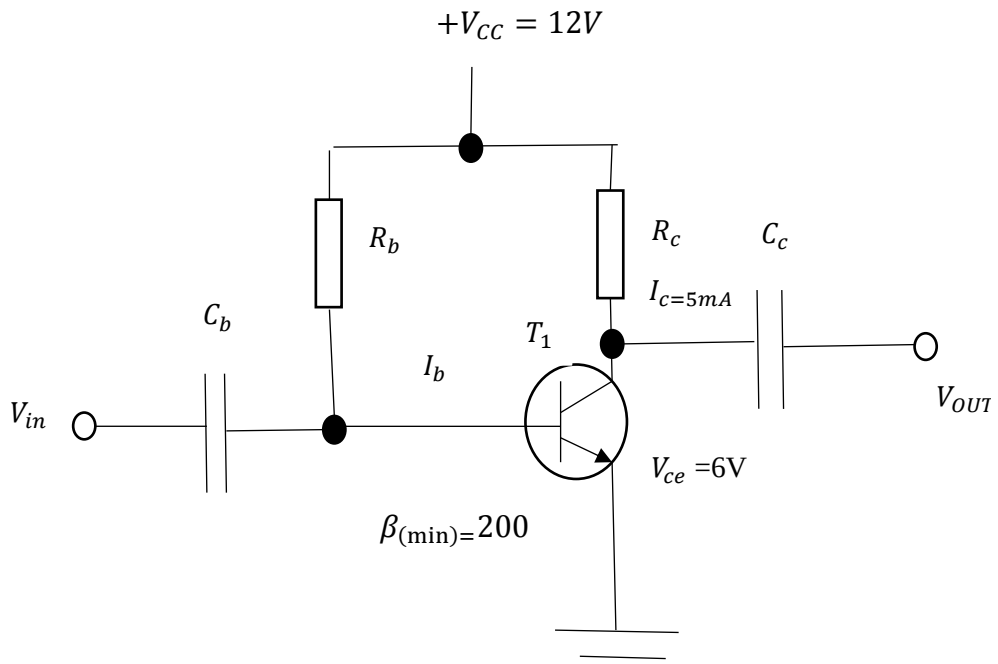


FIGURE 3.4 Fixed bias without R4

Solution:

$$R_C = \frac{V_{CC} - V_{ce}}{I_c} = \frac{12 - 6}{0{,}005} = 1{,}2k\Omega$$

$$I_b = \frac{I_c}{\beta} = \frac{0{,}005}{200} = 25\mu A$$

$$R_b = \frac{V_{cc} - V_{be}}{I_B} = \frac{12 - 0{,}7}{25X\ 10^{-6}} = 452k\Omega$$

- ***If the circuit uses emitter stabilisation***, the voltage drops V_c and current I_e must be brought into account
- It can be accepted that V_e must normally be equivalent to one-tenth of the supply voltage and I_e approximately equal to I_c
- To calculate the value of C_eThe lowest frequency of the input signal must be known

Furthermore, the reactance X_C must be at least 1/10th of the resistor R_e, which means that:

$$C \geq \frac{10}{2\pi f R_e} \quad \left(from\ C = \frac{1}{2\pi f X_c}\right)$$

3.5 BIASSING WITH THE AID OF A VOLTAGE DIVIDER

In the previous amplifier, the values of the bias circuit ***and voltage are dependent on the current gain β of the transistor.*** Because ***the value of β is temperature-sensitive***, especially at silicon transistors, and usually cannot be determined exactly, it is necessary to supply a DC–biased circuit ***independent*** of the transistor β. A circuit of such is shown in ***FIGURE 3.5*** and uses the voltage divider network $R_{b1}\ and\ R_{b2}$.

3.6 DESIGN OF A VOLTAGE DIVIDER BIAS AMPLIFIER

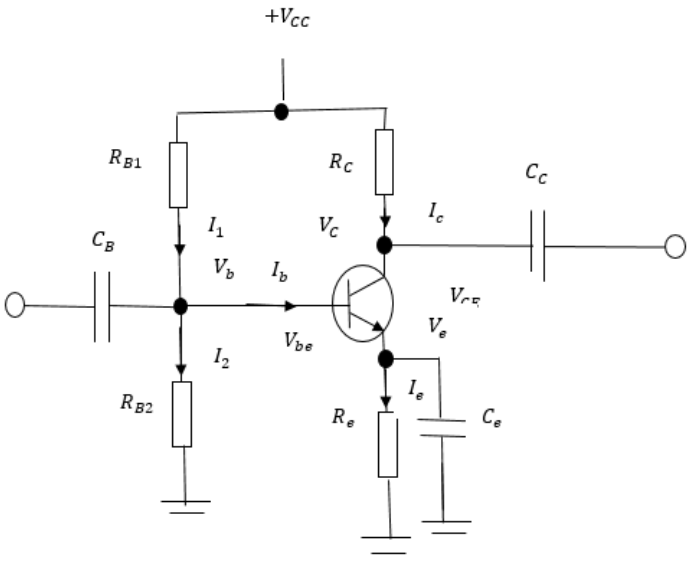


FIGURE 3.5 DC- Bias by means of a voltage divider network

- **The voltage V_b** $is\ calculated\ by\ V_b = V_e + V_{be}$

- The value of the base-emitters impedance (Z_1)is high resistance and greater than R_{b2}. Therefore if Z_1 >> R_{b2}, then $I_{1=}I_2, and\ I_1 \gg\ I_{b.}$
- The equations for the calculation $R_{b1}\ and\ R_{b2}$ are derived as follows: to let the circuit work thoroughly, $I_1\ must = I_2$ and satisfy that, ***R_{b2} must be ≤1/10 (β R_e)***
- $Because\ \boldsymbol{V_b} = \frac{\boldsymbol{R_{b2}}}{\boldsymbol{R_{b1}+ R_{b2}}}\ (\boldsymbol{V_{cc}})\quad ,\quad \boldsymbol{R_{b1}} = \frac{\boldsymbol{R_{b2}(V_{cc}- V_b)}}{\boldsymbol{V_b}}$

Example 3.2

Calculate the values of $R_{e,}\ C_{e,}\ R_{c,}\ \ R_{b2}\ and\ R_{b1}$ in FIGURE 3.6

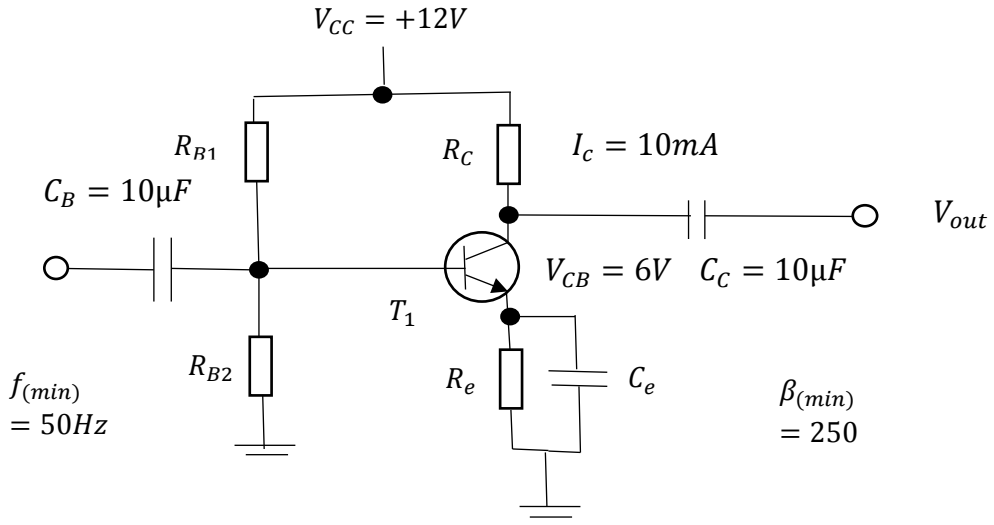


FIGURE 3.6 For Example 3.2

$$V_e = \frac{1}{10}(V_{cc}) = \frac{1}{10}(12) = 1{,}2V$$

$$R_e = \frac{V_e}{I_C} = \frac{1{,}2}{0{,}01}\ 120\Omega$$

$$C_e \geq \frac{10}{2\pi f R_e} = \frac{10}{(2\pi)(50)(1200)}\ 265\mu F$$

$$R_c = \frac{V_{cc}-V_{ce}-V_e}{I_e} = \frac{12-6-1{,}2}{10mA} = 480\Omega$$

$$R_{b2} = \frac{1}{10}(\beta\ R_e) = \frac{1}{10}(250)(120) = 3k\Omega$$

$$V_b = V_e + V_{be} = 1{,}2 + 0{,}7 = 1{,}9V$$

$$R_{b1} = \frac{R_{be}\,(V_{cc} - V_b)}{V_b} = \frac{(3000)(12-1{,}9)}{1{,}9} = 15{,}947\text{k}\Omega$$

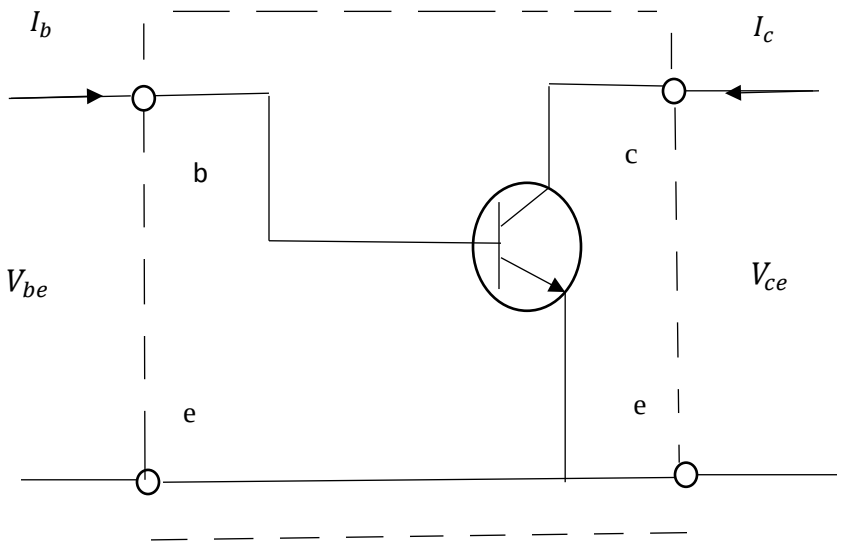
3.7 INTRODUCTION TO HYBRID PARAMETERS

The ***term hybrid is derived from the term "mixed origin"*** and references the mixing of relations in terms of the parameters of transistors. This relation mainly consists of the transistor impedance, current, and voltage characteristics. Therefore, hybrid parameters (h- parameters). As soon as a transistor amplifier stage is designed for certain DC – needs, ***the following step is determining the AC – properties of the circuit. These properties concern the voltage gain, current gain input impedance and output impedance of the circuit.***

- It is necessary to determine the impedance and gain properties when two circuits must match
- ***Mis-matched circuits cause signals to be distorted and devices to be overdriven***. The first step is to reduce the circuit to its AC- format, a system with two input terminals, the so-called "black box" or two-port device. The purpose is to eliminate the DC–circuit so that an overall small-signal AC- analysis can be done.

3.8 THE TRANSISTOR AS A TWO-PORT DEVICE

FIGURE 3.7depicts how a 4- terminal network can replace the transistor in a common emitter configuration. The input circuit is presented by Thevenin's equivalent circuit and the output circuit by a Norton equivalent circuit.



A.) Transistor as a two – port device

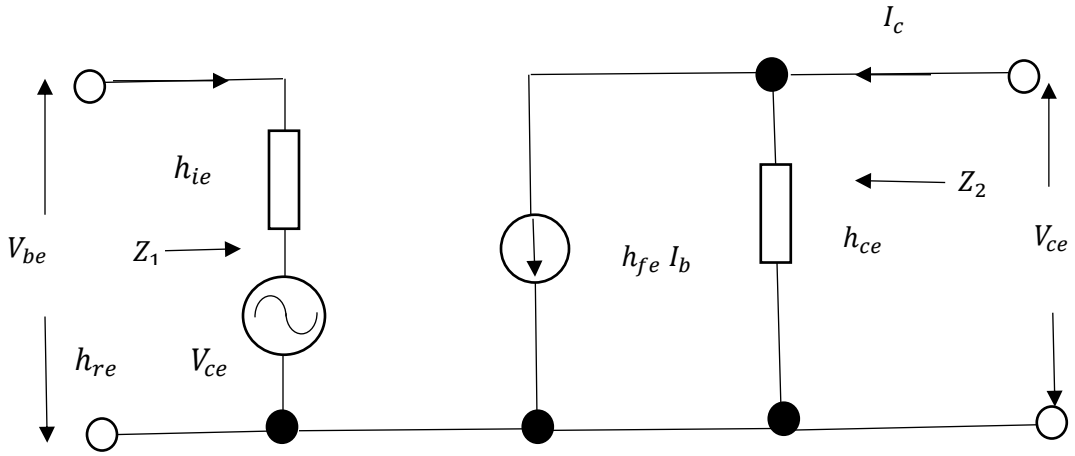


FIGURE 3.7 B.) h- Equivalent circuit

The two general equivalences that apply to the common-emitter configuration are

$$V_{be} = (h_{ie})(ib) + (h_{re})(V_{ce})$$

$$ic = (h_{fe})(ib) + (h_{oe})(V_{ce})$$

$Where\ h_{ie} = \frac{\Delta V_{be}}{\Delta I_b}\ (input\ impedance)(\Omega), V_{ce} = constant$

$h_{re} = \frac{\Delta V_{be}}{\Delta V_{ce}}\ (inverse\ voltage\ gain),\ ib = constant$

$h_{fe} = \frac{\Delta I_c}{\Delta I_b}\ (Foward\ current\ gain), V_{ce} = constant$

$h_{oe} = \frac{\Delta i_c}{\Delta V_{ce}}\ (Output\ admittance)(siemens\ \Omega^{-1}), ib = constant$

$* Admittance\ is\ the\ parallel\ impedance\ \frac{1}{z}\ and\ is\ measured\ in\ siemens(\Omega^{-1})$

All values of h- parameters are usually supplied in the manufacturer's data information sheets. The technician must interpret and apply this information practically. **FIGURE 3.8** shows how devices are mismatched or matched.

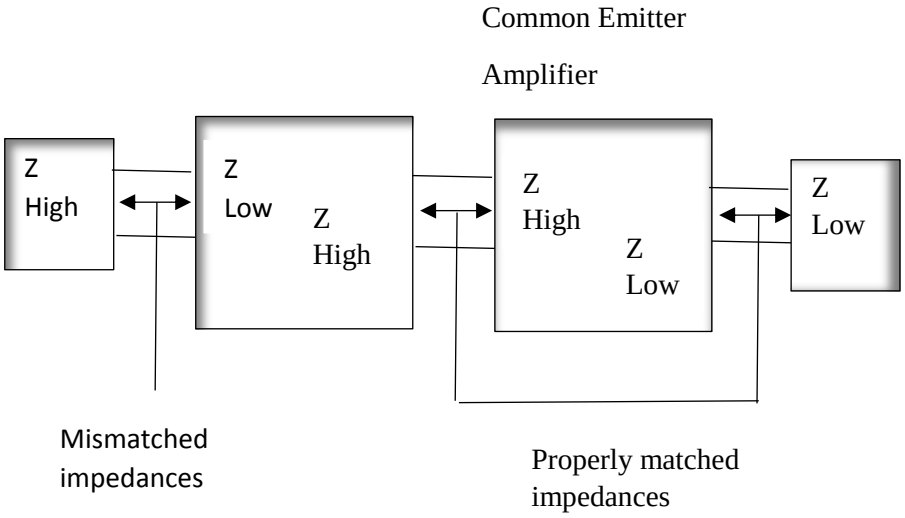
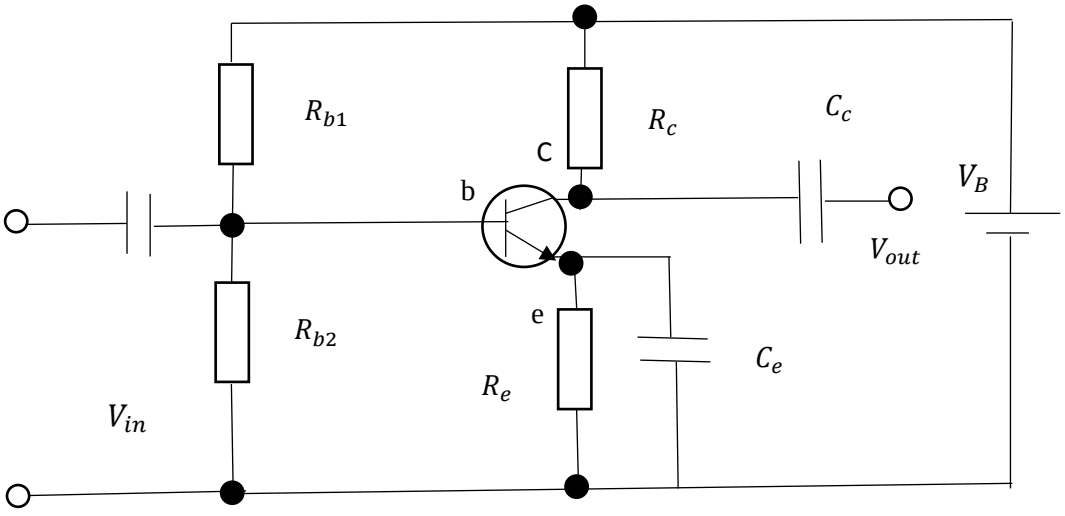


FIGURE 3.7 Mis and properly matched impedances

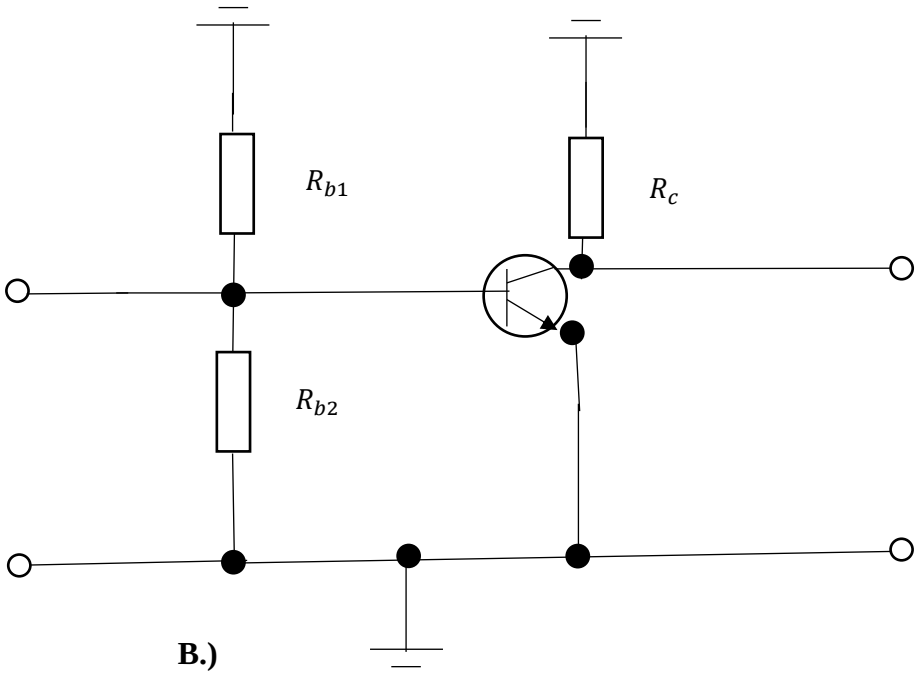
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3.9 EQUIVALENT CIRCUITS AND CALCULATIONS OF COMMON EMITTER AMPLIFIERS

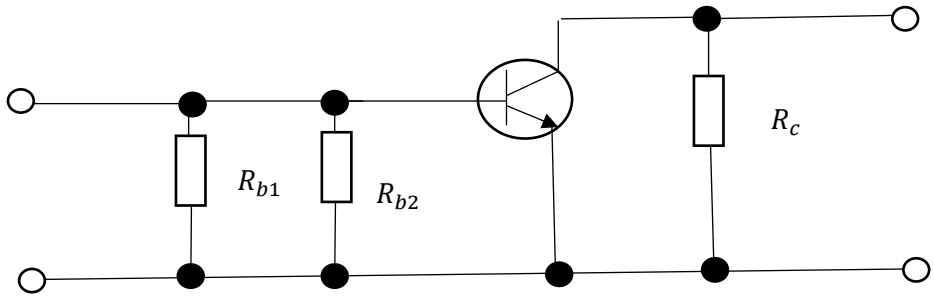
Capacitors and batteries have the properties to produce low reactance at high-frequency signals and can thereby be seen as a short circuit in AC circuits. **FIGURE 3.10** depicts how a common emitter amplifier develops to an equivalent circuit.



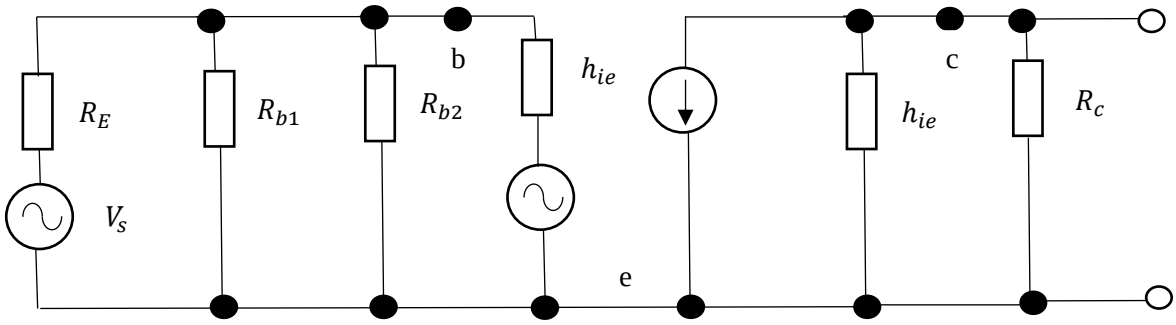
A.)



B.)



C.)



D.)

FIGURE 3.10 the voltage divider bias amplifier and its equivalent h-circuit

The resistor R_s represents any resistor in series with the source and the internal resistance of the source in total. Refer to the summary of the precise and approximate equations below. ***These comparisons can be derived from Ohm's and Kirchhoff's laws.***

Voltage Gain:

$$\textbf{Precise}: \qquad A_V = \frac{(-h_{fe})(Z_L)}{[(h_{ie}) + (h_{ie})(h_{oe}) - (h_{fe})(h_{re})](Z_{L)}}$$

$$Approximate: \ A_V = \frac{(-h_{fe})(Z_L)}{h_{ie}}$$

Current Gain:

$$\textbf{Precise}: \qquad A_i = \left(\frac{h_{fe}}{1+(h_{oe})(Z_L)}\right)\left(\frac{R_{bT}}{R_{bT}+ Z_1}\right)\left(\frac{R_C}{R_C+R_L}\right) = \frac{h_{fe}}{1+ (h_{oe})(Z_L)}$$

Without the current divider rule

$$Approximate: \ A_i = h_{fe}$$

Input impedance:

$$\textbf{Precise}: \qquad Z_I = h_{ie} - \frac{(h_{fe})(h_{re})(Z_L)}{1 + (h_{oe})(Z_L)}$$

$$Approximate: \ Z_I = h_{ie}$$

Output impedance:

$$\textbf{Precise}: \qquad Z_2 = \frac{1}{h_{oe} - \frac{(h_{fe})(h_{re})}{h_{ie+}\, R_S}}$$

$$Approximate: \ Z_2 > \frac{1}{h_{oe}}$$

Power Gain:

$$\textbf{\textit{Precise}}: \quad A_p = \frac{(A_i)(R_L)}{R_s}$$

$$Approximate: \ A_p = \frac{(h^2{}_{fe})(R_L)}{h_{ie}}$$

ANNOTATION: $R_S = Thevinin - impedance$

If the base and collector resistors are taken into consideration, then the following applies:

$Z_0 = Z_L // Z_2$, $Where\ Z_L =$ R_C or $Z_L = R_C$ // R_L (if R_L is given) and

$Z_i = R_{b1} // R_{b2} // Z_1$ $for\ a\ voltage\ divider\ network$ **or**

$Z_i = R_b // Z_1$ $for\ a\ fixed\ bias\ network.$

Precise or approximate methods can therefore calculate all gains and impedances.

Example 3.3

Determine the following values of the fixed bias amplifier in **FIGURE 3.11** using the precise method.

A_i, A_v, Z_i, Z_o, A_p

$h_{ie} = 1{,}2k$; $h_{re} = 2X10^{-4}$; $h_{fe} = 60$; $h_{oe} = 20\mu A = 20\mu\, Siemens$

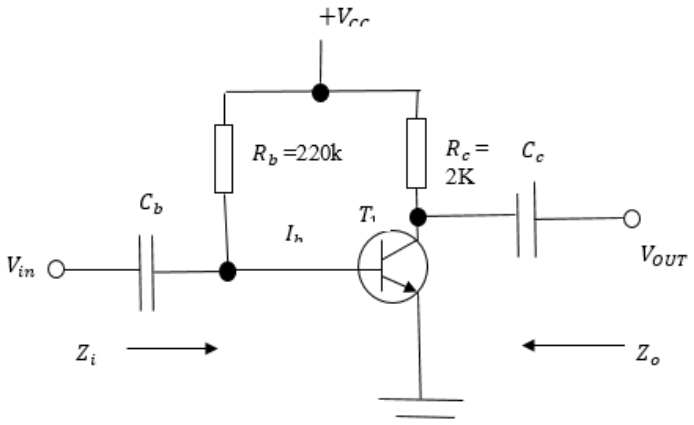


FIGURE 3.11 Circuit for Example 3.3

Solution:

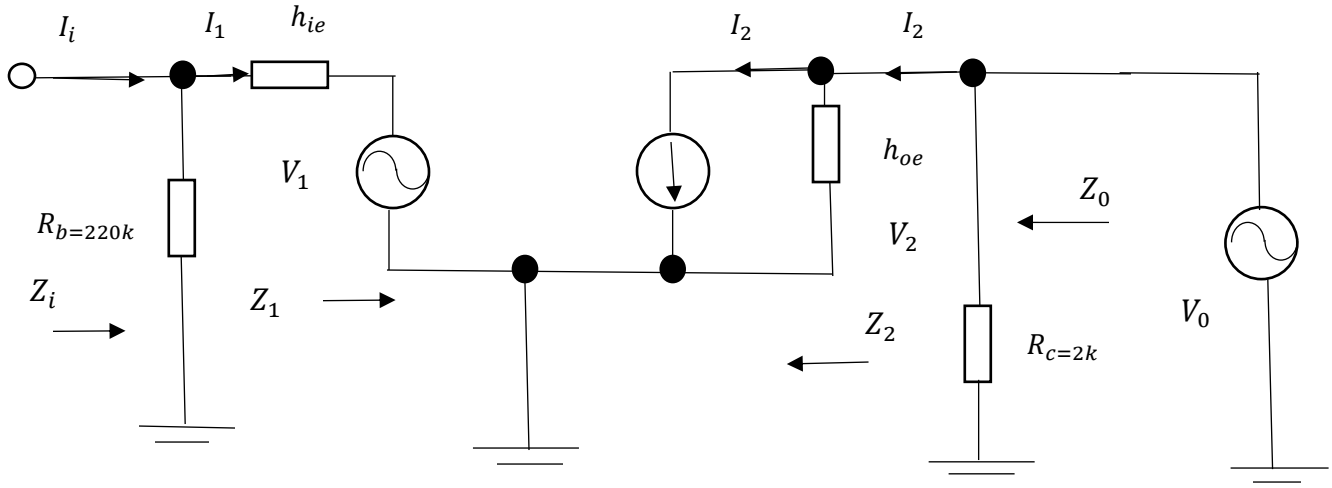


FIGURE 3.11.1 H – equivalent for small signal AC - analysis

A.) $Z_1 = h_{ie} - \frac{(h_{fe})(h_{re})(Z_L)}{1+(h_{oe})(Z_L)} = 1200 - \frac{(60)(2X10^{-4})(2000)}{1+(20X10^{-6})(2000)} = 1{,}176k\Omega$

$R_b \gg Z_1, I_i = I_1$,$\boldsymbol{Therefore\ A_i = \frac{I_O}{I_1} = \frac{I_O}{I_i} = \frac{I_2}{I_1}} = \left(\frac{h_{fe}}{1+(h_{oe})(Z_L)}\right)\left(\frac{R_b}{R_b+Z_1}\right) =$
$\left(\frac{60}{1+(20\ X\ 10^{-6})(2000)}\right)\left(\frac{220000}{220000+1176}\right) = 57{,}3$

B.) $\boldsymbol{A_V} = \frac{V_O}{V_i} = \frac{V_2}{V_1} = \frac{(-h_{fe})(Z_L)}{[(h_{ie})+(h_{ie})(h_{oe})-(h_{fe})(h_{re})](Z_L)}$

$= \frac{(-60)(2000)}{(1200)+(1200)(20X10^{-6})-(60)(2X10^{-4})(200)} = -98$

C.) $\boldsymbol{Z_i} = R_b // Z_1 = = \frac{220k\Omega}{1{,}176k\Omega} = 1{,}169k\Omega$

$\boldsymbol{Z_O} = Z_L \,//Z_2 = R_C \,//\, Z_2$, where $Z_2 = \; Z_2 = \frac{1}{h_{oe} - \frac{(h_{fe})(h_{re})}{h_{ie+}\, R_S}} = \frac{1}{(20X10^{-6}) - \frac{(60)(2X10^{-4})}{1200+0}}$

$=100k\Omega$

$Z_O = R_C \,//\, Z_2 = 2k\Omega \,//\, 100k\Omega \; = 1{,}96k\Omega$

$D.)\; \boldsymbol{A_{P=}}(-A_V)(A_i) = -(-98)(57{,}3) = 5615{,}4$



Example 3.3.1

Repeat Example 3.3 now using the approximate equations, and compare these values to the Precise Values



Solution:

A.) $\boldsymbol{A_i} = \; h_{fe} = 60$

B.) $\boldsymbol{A_V} = \frac{(-h_{fe})(Z_L)}{h_{fe}} = \frac{(-60)(2000)}{1200} = \; -100$

C.) $\boldsymbol{Z_i} = \; R_b//Z_I = \; h_{ie} = 1{,}2k\Omega$

D.) $Z_2 = \frac{1}{h_{oe}} = \frac{1}{20X10^{-6}} = 50k\Omega$, $\boldsymbol{Z_o} = \; R_c//Z_2 = 2k\Omega \,//\, 50k\Omega = 1{,}923k\Omega$

E.) $\boldsymbol{A_p} = (-A_v)(A_i) = -(-100)(60) = 6000$

Now we compare our Precise Values to our Approximate values



	Precise	Approximate
$\boldsymbol{A_V}$	-98	-100
$\boldsymbol{Z_i}$	1,169kΩ	1,2kΩ
$\boldsymbol{Z_o}$	1,96kΩ	1,923kΩ
$\boldsymbol{A_p}$	5615,4	6000
$\boldsymbol{A_i}$	57,3	60

Example 3.4

Calculate the following values of the voltage divider amplifier in **FIGURE 3.12** using the precise method.

A.) A_V
B.) A_i
C.) Z_i
D.) Z_o
E.) A_p

$h_{ie} = 1{,}2k\ ;\ h_{re} = 2X10^{-4}; h_{fe} = 100\ ;\ h_{oe} = 20\mu A$

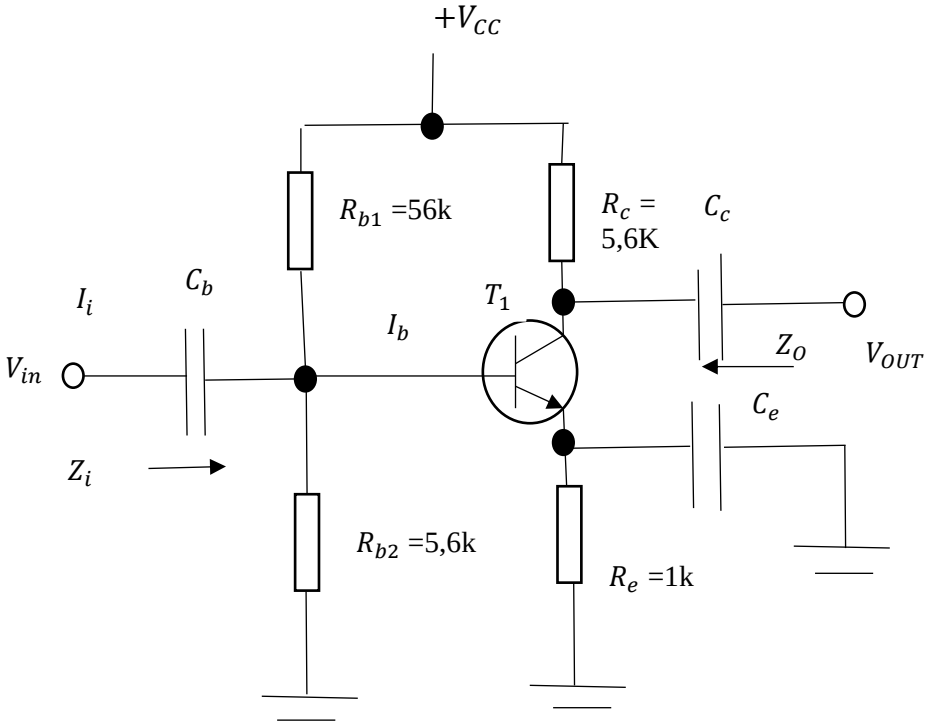


FIGURE 3.12 Circuit for Example 3.4

Solution:

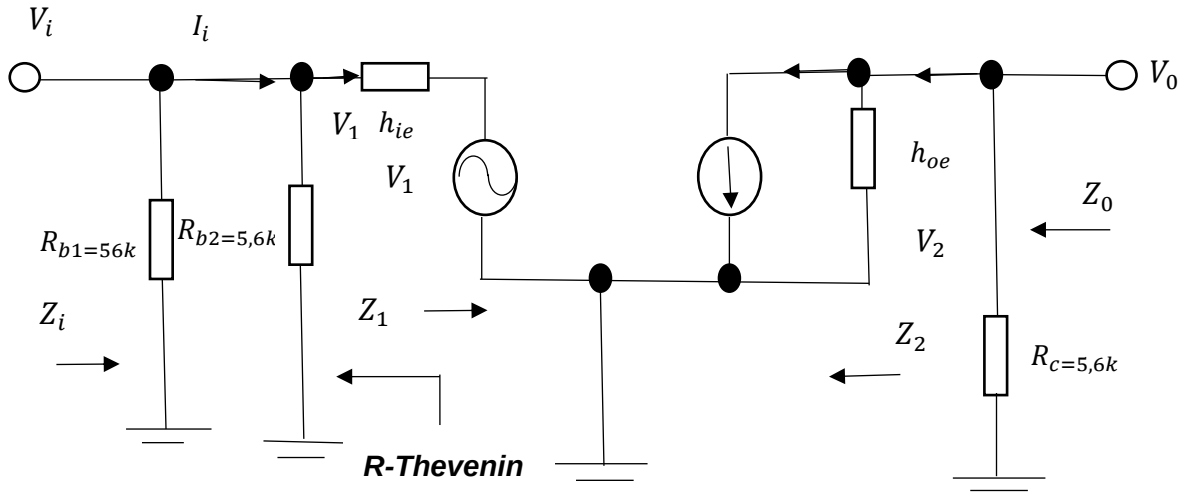


FIGURE 3.13.1 H – Equivalent circuit for small signal AC- analysis

A.)

$$\boldsymbol{A_v} = \frac{V_O}{V_i} = \frac{V_2}{V_1} = \frac{(-h_{fe})(Z_L)}{[(h_{ie}) + (h_{ie})(h_{oe}) - (h_{fe})(h_{re})](Z_{L)}}$$

$$= \frac{(-100)(5600)}{1200 + (1200X10^{-6}) - (100)(2X10^{-4})(5600)}$$

$$= -458{,}115$$

B.) Z_1 *is required for the calculation of* I_i and I_I

$$Z_1 = h_{ie} - \frac{(h_{fe})(h_{re})(Z_L)}{1 + (h_{oe})(Z_L)}$$

$$1200 - \frac{(100)(2X10^{-4})(5600)}{1 + (20X10^{-6})(5600)}$$

=1,099kΩ

Since $R_{b1}//R_{b2}$ is not greater than Z_1 the relation between I_i *and* I_I must be calculated according to the current divider rule:

$$I_I = \frac{R_{bT}}{R_{bT} + Z_1}(I_i)$$

$$\frac{I_I}{I_i} = \frac{R_{bT}}{R_{bT} + Z_1}$$

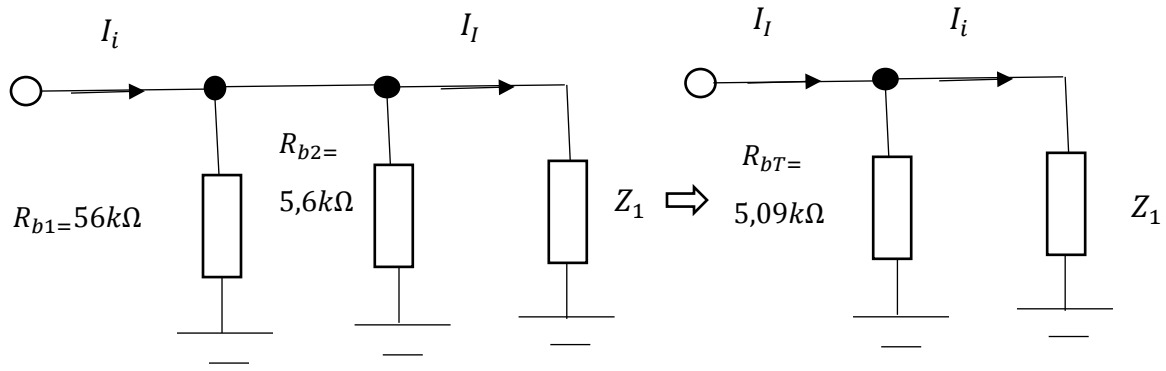


FIGURE 3.14

$$\frac{I_I}{I_i} = \frac{R_{bT}}{R_{bT} + Z_1} = \frac{5{,}09k\Omega}{5{,}09k\Omega + 1{,}099k\Omega} = 0{,}822$$

$$\boldsymbol{A_i} = \frac{I_o}{I_i} = \frac{I_2}{I_i} = \left(\frac{I_2}{I_I}\right)\left(\frac{I_I}{I_i}\right) = \left(\frac{h_{fe}}{1 + h_{oe}(Z_L)}\right)\left(\frac{R_{bT}}{R_{bT} + Z_1}\right) = \left(\frac{100}{1 + (20X10^{-6})(5600)}\right)(0{,}822) = 73{,}9$$

C.) $\boldsymbol{Z_1} = 56k\Omega \,//\, 5{,}6k\Omega \,//\, Z_1$
$= 5{,}09 \,//\, 1{,}099$
$= 0{,}903k\Omega$

D.) R_s is required to determine Z_2=$R_s = R_{TH}$ and R_{TH} is indicated in **FIGURE 3.13.1** to calculate R_{TH} The AC source V_i must be short-circuited as well, and thereby $R_S = R_{TH} = 0$.

$$\mathrm{Z_2} = \frac{1}{\mathrm{h_{oe}} - \frac{(\mathrm{h_{fe}})(\mathrm{h_{re}})}{\mathrm{h_{ie+}\,R_S}}}$$

$$= \frac{1}{(20X10^{-6}) - \frac{(100)(2X10^{-4})}{1200 + 0}} = 300k\Omega$$

$$\boldsymbol{Z_o} = Z_2 // R_c$$

$$= 300k\Omega \,//\, 5{,}6k\Omega$$

$$= 5{,}49k\Omega$$

E.) $\boldsymbol{A_p} = (-A_v)(A_i) = -(-458{,}115)(73{,}9) = 33854{,}69$

Example 3.5

Repeat Example 3.4 using approximate equations and compare the results.

Solution:

A.) $\boldsymbol{A_V} = \frac{(-h_{fe})(Z_L)}{h_{fe}} = \frac{(-100)(5600)}{1200} = -466$

B.) $I_O = (h_{fe})(I_b), current\ divider\ rule\ I_b = \frac{R_{b2}}{R_{b2}+h_{ie}}(I_i) = \frac{I_b}{I_i} = \frac{5{,}6k\Omega}{5{,}6k\Omega+1{,}2k\Omega} = 0{,}823$

$Note:\ R_{b1}: R_{b2} = 10:1, therefore\ R_{b1}\ can\ be\ ignored\ \boldsymbol{A_i} = \frac{I_o}{I_i} = (h_{fe})\left(\frac{I_b}{I_c}\right)$

$= (h_{fe})\left(\frac{R_{b2}}{R_{b2}+h_{ie}}\right) = (100)(0{,}823) = 82{,}3$

C.) $\boldsymbol{Z_i} = R_{bT}\ //\ Z_1 = 5{,}09k\Omega //\ h_{ie} = 5{,}09k\Omega\ //\ 1{,}2k\Omega = 971\Omega$

D.) $Z_2 > \frac{1}{h_{oe}} = \frac{1}{20X10^{-6}} = 50k\Omega$; $\boldsymbol{Z_o} = R_c // Z_2 = 5{,}6k\Omega\ //\ 50k\Omega = 5{,}035\ k\Omega$

E.) $\boldsymbol{A_p} = (-A_v)(A_i) = -(-466)(82{,}3) = 38351{,}8$

	Precise	Approximate
A_V	-458,115	-466
A_i	73,9	82,3
Z_i	0,903kΩ	971Ω
A_p	33854,69	38351,8
Z_o	5,49kΩ	5,035kΩ

3.10 COMMON BASE AMPLIFIERS

The common base amplifier has a high voltage gain but less than one current gain. The equations used at the common emitter amplifier are applicable here as well. Only the footnote is now replaced by a b ($h_{ie}\ now\ becomes\ h_{ib}$.)

Example 3.6

Refer to **FIGURE 3.15** and determine the following values:

A_i, A_v, Z_i, Z_o

$h_{ib} = 45\Omega$; $h_{rb} = 200X10^{-6}$; $h_{ob} = 0{,}4\mu A/V$; $h_{fb} = -0{,}98$

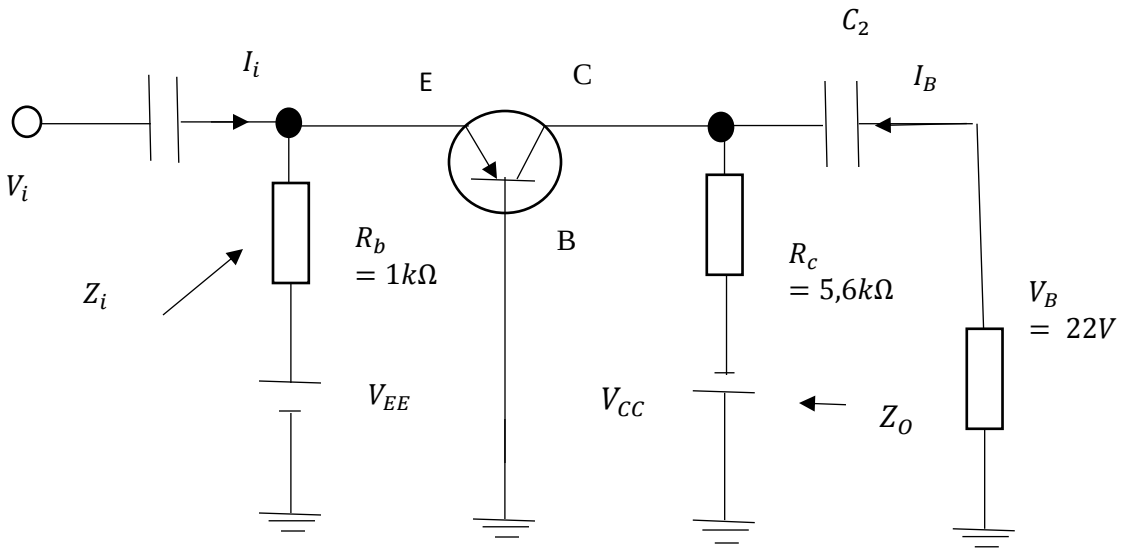


FIGURE 3.15 Circuit diagram for example 3.5

In **FIGURE 3.16,** below, all capacitors and DC supplies are replaced by short circuits to give you the h-equivalent of **FIGURE 3.15.**

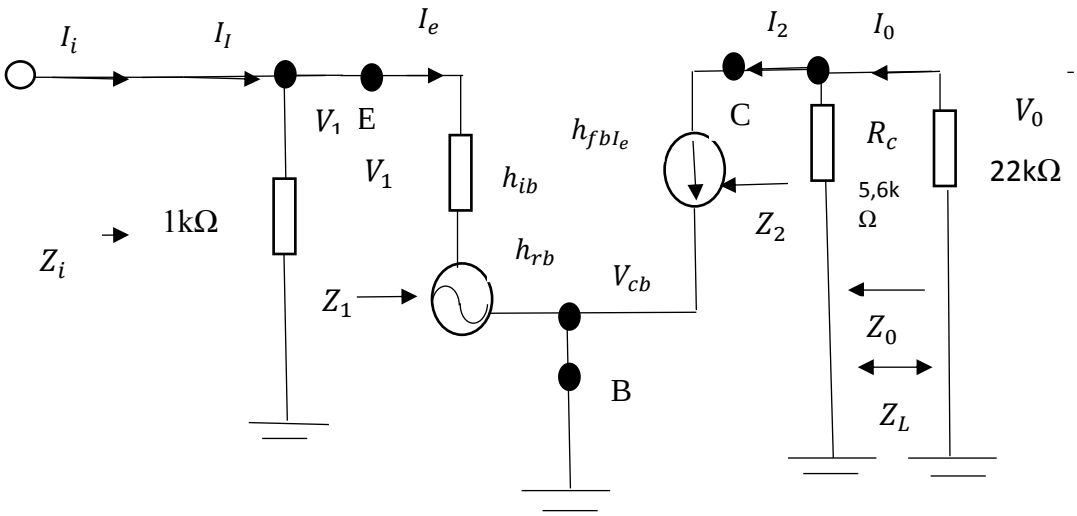


FIGURE 3.16 H – Equivalent circuit for small signal AC-analysis

A.) The equivalent load : $Z_L = \frac{R_C}{R_L} = \frac{5{,}6K\Omega}{22K\Omega} = 4{,}464k\Omega$

$$Z_L = h_{ib} - \frac{(h_{fb})(h_{rb})(Z_L)}{1+(h_{ob})(Z_L)} = 45 - \frac{(-0{,}98)(200X10^{-6})(4{,}464\ X\ 10^3)}{1+(0{,}4\ X\ 10^{-6})(4{,}464X10^3)} = 45{,}8\Omega$$

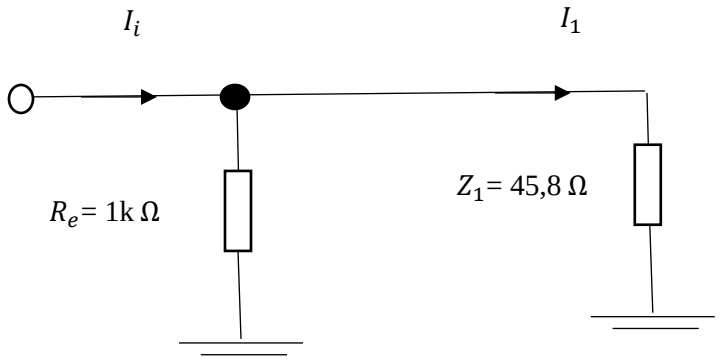


FIGURE 3.17 – Relation between I_i and I_1

According to the current divider rule

$$I_1 = \frac{1k\Omega}{1k\Omega\ 45{,}8\Omega}\ (I_i)$$

$$\frac{I_1}{I_i} = \frac{1000}{1045{,}8} = 0{,}96$$

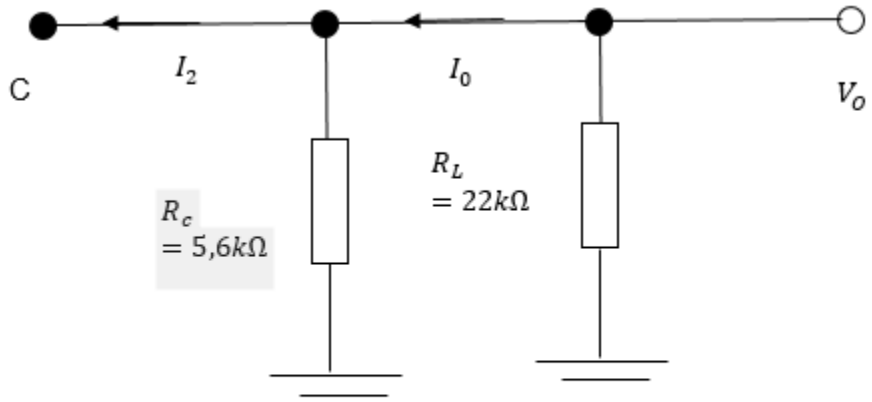


FIGURE 3.18 – Relation between I_0 and I_2

According to the current divider rule

A.) $I_0 = \frac{5,6K\Omega}{5,6k\Omega+22k\Omega}(I_2)$

$\frac{I_0}{I_2} = \frac{5600}{27600} = 0,2$

$A_i = \frac{I_0}{I_i} = \left(\frac{h_{fe}}{1+(h_{ob})(Z_L)}\right)\left(\frac{R_e}{R_e + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$

$= \left(\frac{-0,98}{1+(0,4X10^{-6})(4,464X10^3)}\right)(0,96)(0,2) = -0,188$

B.) $A_v = \frac{V_0}{V_i} = \frac{V_2}{V_1}$

$= \frac{(-h_{fe})(Z_L)}{h_{ib}+(h_{ib})(h_{ob})-(h_{fb})(h_{rb})(Z_L)} = \frac{-(-0,98)(4,464X10^3)}{45-(45)(0,4X\,10^{-6})(-0,98)(200X10^{-6})(4,464X10^3)} = 95,5$

C.) $Z_i = R_c \,//\, Z_I$

$= 1k\Omega \,//\, 45,8\Omega$

$= 43,8\Omega$

D.) $Z_2 = \frac{1}{h_{ob}-\frac{(h_{fb})(h_{rb})}{h_{ib+}R_S}}$

$= \frac{1}{(0,4X10^{-6})(\frac{(-0,98)(200X10^{-6})}{45+0}}$

$= 210k\Omega$

$Z_o = R_c//Z_2$

$= 5,6k\Omega \,//\, 210k\Omega = 5,45k\Omega$



Example 3.7

Repeat Example 3.6 using the approximate method. The equivalent circuit in FIGURE 3.16 is simplified further in FIGURE 3.19



Solution:

A.) According to the current divider rule:

$I_o = \frac{5,6k\Omega}{5,6k\Omega+20k\Omega}(I_2)$

$\frac{I_o}{I_2} = 0,2$

$$A_i = (h_{fb})\left(\frac{R_c}{R_c + R_L}\right)$$
$$= (-0{,}98)(0{,}2)$$
$$= -0{,}196$$

B.) $A_v = \frac{(-h_{fb})(Z_L)}{h_{ib}}$
$$= \frac{-(-0{,}98)(4{,}464k\Omega)}{45}$$
$$= 97{,}2$$

C.) $Z_i = h_{ib} = 45\Omega$

D.) $Z_o = R_e = 5{,}6k\Omega$

3.11 DISTORTION OF A COMMON EMITTER AMPLIFIER

The amplified output signal of an amplifier should be alike to the input signal. If the output signal does not favor the layout of the input signal, we say that the output signal is distorted.

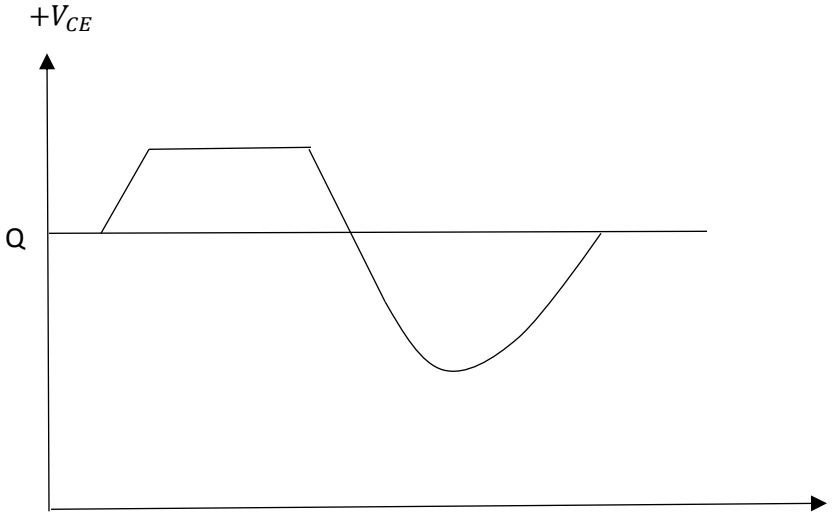
There are a few factors which cause this distortion. These include:

- The amplitude of the input signal is too huge
- The work point of the circuit is in error
- The load and circuit output are not adapted
- The input characteristics are not linear

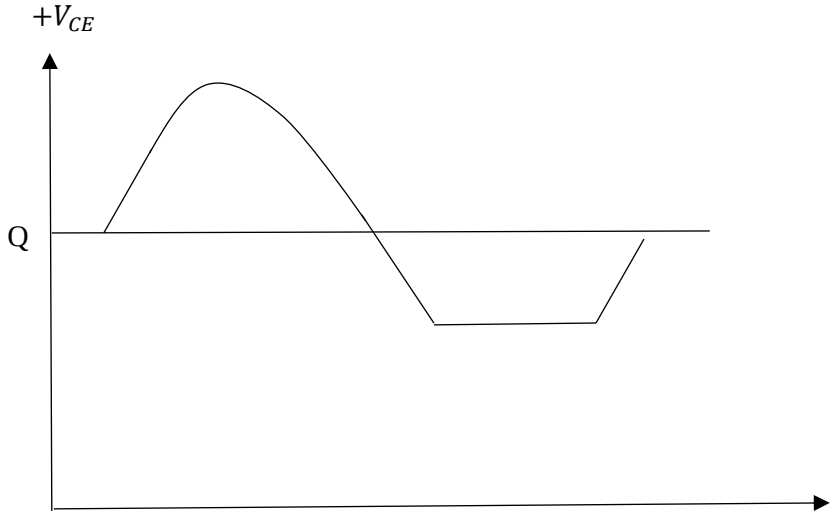
Only a thorough design which considers the above factors will assure an efficient amplifier.

3.12 AMPLITUDE DISTORTION

An output signal with a similar but amplified amplitude will be produced when an input signal is applied to an amplifier. If the input amplitude is increased, the output amplitude will increase correspondingly. However, there is a limitation on the amplification factor. ***An input amplitude that is too big will drive the transistor to either saturation point or cut-off point, which causes chopping off the upper part of the signal, the lower part of the signal, or both.***

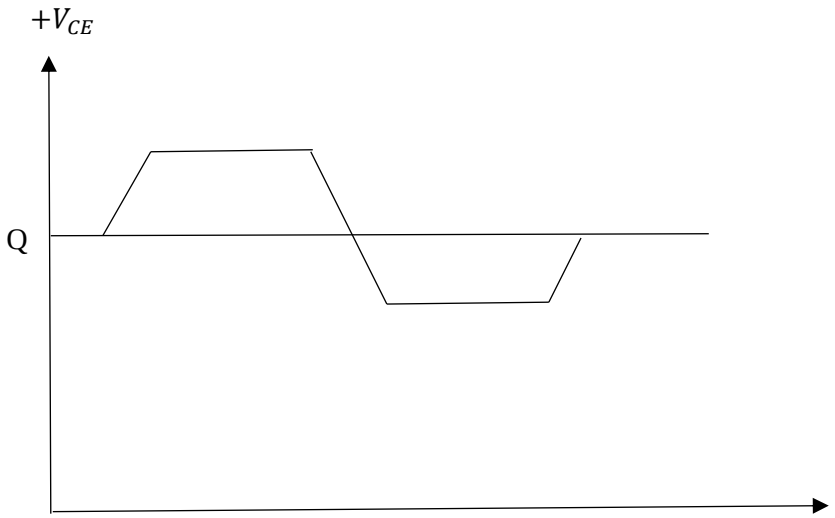


i.) Chopping because of saturation



ii.) Chopping because of cut-off

FIGURE 3.19



iii.) Chopping because of over driven input

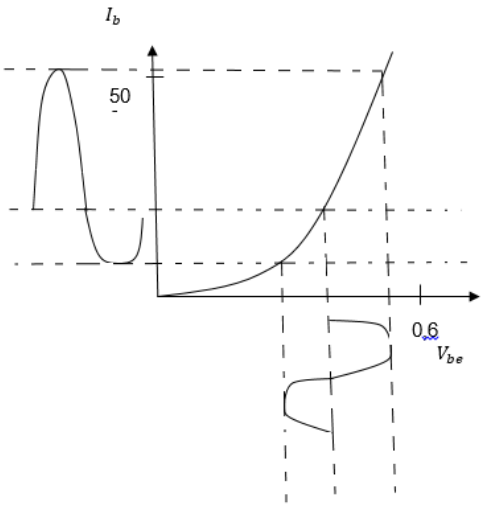
FIGURE 3.19

A work point Q chosen too close to the positive supply will result in ***saturation (Figure i).*** If the work point is too close to 0V, ***the transistor will switch off (Figure ii). Figure iii*** shows that the work point is chosen correctly, but the input amplitude is too big and causes symmetrical chopping.

3.13 NON- LINEAR DISTORTION

The non-linearity of the input characteristics distorts the output signal. ***As soon as the base bias V_{be} drops below the cut–off region, the input current I_b is distorted.*** This effect causes a similar distortion in the output current and output voltage.

The input current distortion can be overcome by using a current source at the input of a voltage supply. A signal source with a high output impedance will force a non-distorted signal into the amplifier input.



Self-Check 3

1.) Mention 3 factors which influence the stability of a transistor amplifier.

2.) Calculate the values of $R_c \; and \; R_b$ in Figure 3.4, if β, $V_{cc} = 10V, I_c = 12mA \; and \; V_{ce} = 8V$. Assume that $V_{be} = 0{,}6V$

3.) Calculate the values $R_{e,} C_e, and \; R_b$ of a fixed bias amplifier in Figure 3.5 if β=280V,$V_{cc} = 15V, I_c = 6mA, V_{ce} = 5V, V_{be} = 0{,}7V, R_c = 1416{,}6\Omega \; and \; f = 50Hz$

4.) Draw the h-equivalent for the circuits for Figures 3.4, 3.5 and 3.6

5.) Calculate the current, voltage and power gain as well as the input and output impedances in Question 2 if the $h_{ie} = 1{,}3k\Omega, h_{re} = 2{,}5X10^{-4}, h_{oe} = 30\mu A. V \; and \; that \; h_{fe} = \beta$ Use the precise method.

6.) Repeat the calculations in Question 6 with the aid of the approximate method.

7.) The following values of the common base amplifier are :$R_e = 1{,}2k\Omega, R_c = 8{,}2k\Omega, the \; load = 15k\Omega, h_{rb} = 200X10^{-6}, h_{ob} = 0{,}47\mu. V \; and \; h_{fb} = -0{,}95$. Draw the h-equivalent circuit of the circuit and calculate :

A.) A_i
B.) A_v
C.) Z_i
D.) Z_o

According to

7.1.) The precise method.

7.2.) The approximate method.

8.) Mention 4 factors that can distort an amplifier.

9.) With the aid of sketches, briefly describe:

9.1.) Amplitude distortion

9.2.) Frequency distortion

9.3.) Non-linear distortion

3.1 INTRODUCTION

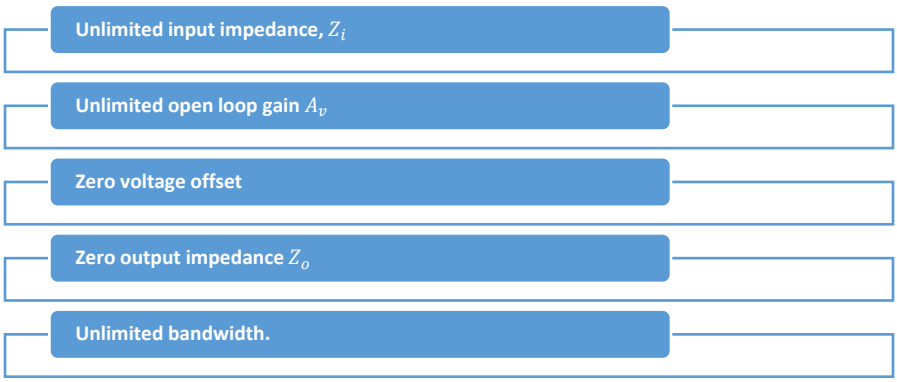
The ***operational amplifier is an integrated circuit with high voltage gain*** and uses external feedback networks for its operation. It is a versatile amplifier that can operate as an amplifier active filter, Signal generator, voltage regulator, etc.

Watch the video for more information on amplifiers:
https://www.youtube.com/watch?v=kiiA6WTCQn0

3.2 THE IDEAL OPERATIONAL AMPLIFIER

An operational amplifier without an external feedback network is an amplifier in the open-loop configuration. In this configuration, the ideal(Theoretically) amplifier ought to **have the following properties:**

In practice, no operational amplifier can satisfy these requirements, and, therefore, no ideal operational amplifier exists.



3.3 BASIC PROPERTIES

Properties of the operational amplifier

- A very high voltage gain is Suitable to amplify very small signals.
- A very low output impedance is not affected easily by other loads.
- A very high input impedance causes almost no current load for circuits.

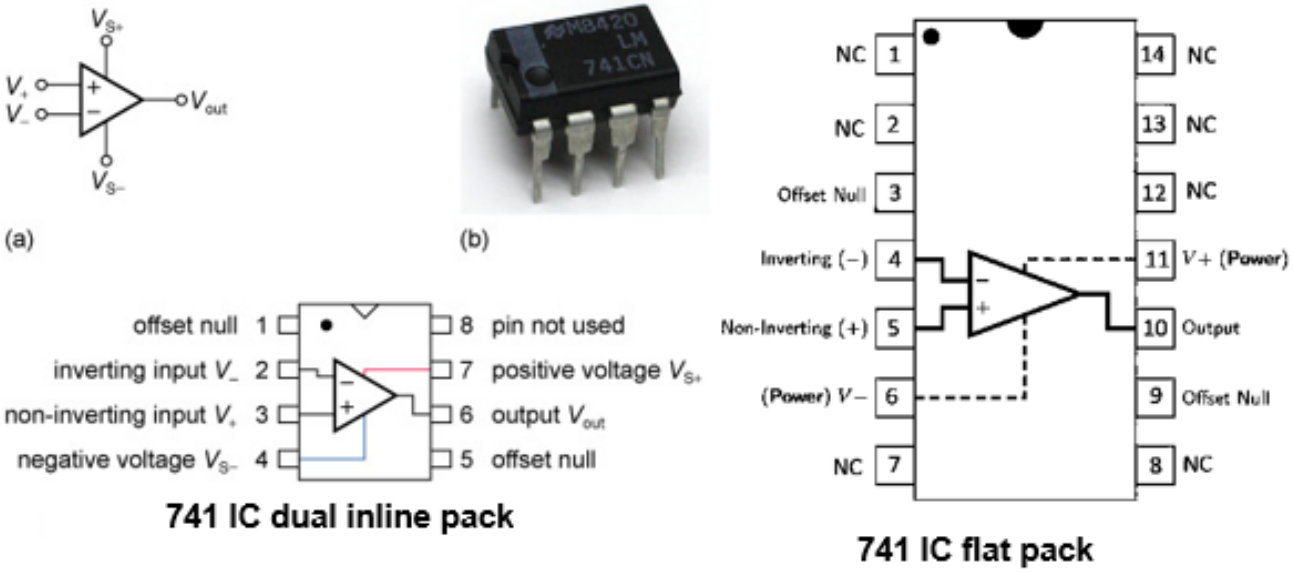
Table 3.1 shows typical properties of the 741 – Operational amplifier.

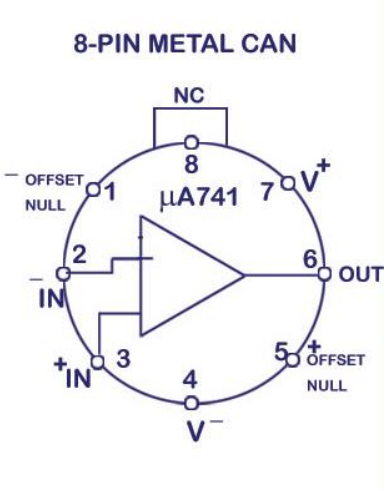
PARAMETER	QUANTITY
$\boldsymbol{Open\ Loop\ voltage\ gain\ A_v}$ $(\boldsymbol{R_L\ \geq 2k\Omega})$ $(\boldsymbol{V_o \pm 10V})$	200 000 15 000
$\boldsymbol{Input\ impedance\ Z_i}$	$1\ M\Omega$
$\boldsymbol{Output\ impedance\ Z_o}$	$150\ M\Omega$
$\boldsymbol{Input\ bias\ current\ I_b}$	$200\mu A$
$\boldsymbol{Output\ current}$	$2{,}8\ mA$
$\boldsymbol{Maximum\ bias\ V_s(max)}$	$\pm 18V$
$\boldsymbol{Maximum\ input\ voltage\ V_i\ (max)}$	$\pm 13V$
$\boldsymbol{Maximum\ Output\ voltage\ V_O\ (max)}$	$\pm 14V$
$\boldsymbol{Power\ dissipation}$	$85\ mW$
$\boldsymbol{Differential\ input\ voltage\ offset\ V_u}$	$2mV$
$\boldsymbol{Common\ mode\ rejection\ ratio\ CMRR}$	$90\ dB$
$\boldsymbol{Frequency\ f_o}$	$1\ MHz$
$\boldsymbol{Switching\ rate\ S}$	$1\ V.\mu\ s$

Table 3.1

3.3.1 Pinout and IC packages of op-amps

The figures below show the op-amp pin layout and both the IC and metal packages.







The SN52741 is featured by its operation across the full military temperature range from 55° to +125°C and the SN72741 for its industrial application in the 0 ° to +70 ° C- region.

APPLICATIONS OF OP-AMPS

The operational amplifier is of the most versatile integrated circuits and can perform so many functions that one can hardly name them. It is applied especially ***in measuring equipment, analog to digital converters, motor speed control circuits, radio equipment, power supplies, research systems,*** etc. Every other function can be performed just by selecting the external components around the integrated circuit. A few of the basic functions follow

3.4 THE DIFFERENTIAL AMPLIFIER CIRCUIT

In **FIGURE 4.2** below, a simplified circuit of the operational amplifier is shown.

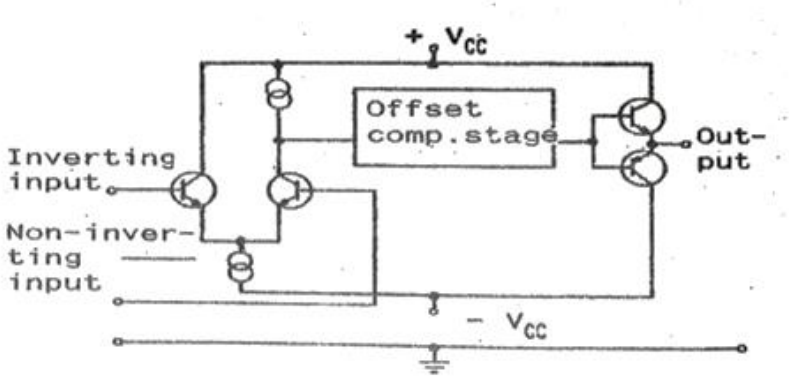


FIGURE 4.2: Simplified circuit of an operational amplifier

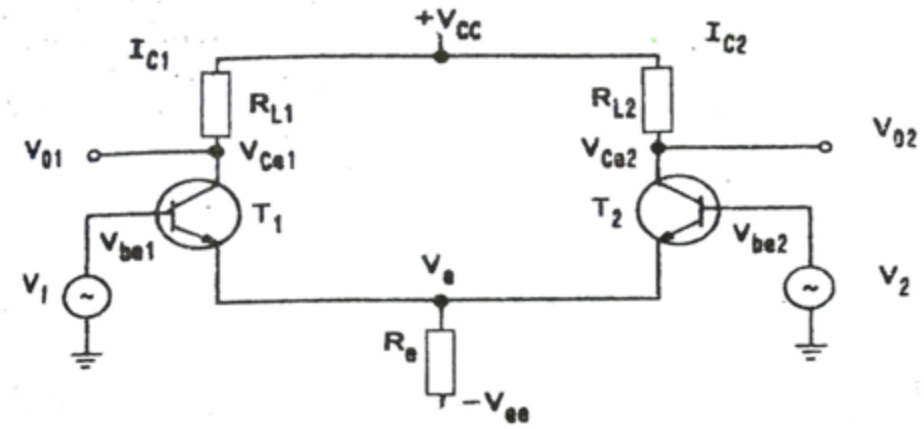


FIGURE 4.3: The differential amplifier

The operation of the differential amplifier is as follows:

- If one of the inputs is zero, say V_2 and V_1 becomes positive about ground, the current through T_1 will increase.
- It will cause the collector potential(V_{cc}) across $T_1$to drop, but the emitter potential (V_e) to increase.
- The increase in the emitter potential decreases the base bias $(V_{be2})\ of\ T_2$ and therefore also the current flow $(I_{c2}) through\ T_2$, the base bias (V_{be2}) of T_2
- Therefore when $V_{o1\ drops,}\ V_{o2}\ rises$ which amplifies a differential or difference between the two input signals. ***Hence the name differential amplifier***.

DRIFT

It is very suitable, especially at low frequency signals and in DC-amplifiers where electrical noise appears as a common mode signal.

Every circuit is subjected to an appearance which is called drift. ***"Drift"*** is a slow variation of the working voltage or current on the output while the input signal is kept at a constant level.

This is caused by various factors:

Various factors Caused by drift
• Temperature variations
• Unstable voltage supply.
• Parameter variations of components
• Aging of components
• Changing load conditions.

Table 3.4

The differential amplifier reduces the drift such as that in figure 4.2 because drift is amplified by both transistors concurrently. The difference between the outputs of the transistors (which is amplified further), will be almost zero.

3.5 COMMON MODE REJECTION RATIO

If two signals, equal in size and phase with each other (like drift), are applied on the inputs, it is known as common mode signals. The differential amplifier has a high "***common mode rejection ratio"*** (CMRR) because the differential output signal is zero for common mode signals.

The differential amplifier has a high gain for differential mode signals mode signals (A_{dm}) and a low gain for common mode signals(A_{cm}). The common mode rejection ratio is therefore calculated by:

$$CMRR = \frac{A_{dm}}{A_{cm}}$$

The higher this value, the more influential the amplifier.
This factor can also be expressed in decibels:

$$CMRR\ (dB) = 20 \log \frac{A_{dm}}{A_{cm}}$$

A disadvantage of this circuit is that the transistor practically shows a disorder and causes a small deviation of bias potential at input terminals. This deviation is known as the ***differential input offset V (±2 mV)*** and can be amplified effectively by the operational amplifier. To eliminate this unwanted offset, most operational amplifiers possess "***the null balance facility.***"

Watch the video for more information on The Null Balance Facility**:**
https://www.youtube.com/watch?v=uYryH28yFAc

Theoretically spoken, every input must 'see' the same Resistance to the ground in critical applications. This means that ***every input will have the same current for the null Condition and therefore no*** output. ***If the design does not allow equal resistance to the ground, a potentiometer*** can compensate for these unequal currents.

FIGURE 3.4 shows how the null potentiometer, usually with a value ***of*** 10 kΩ, must be connected.

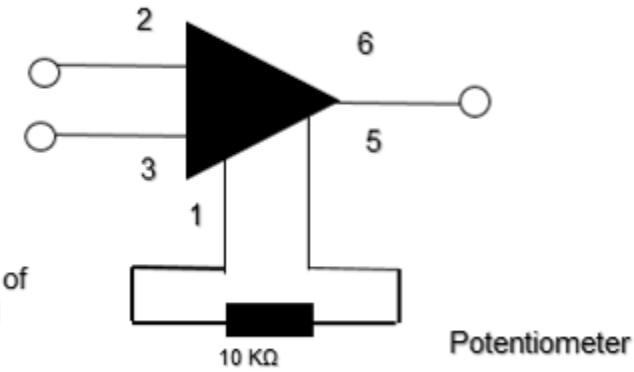


FIGURE 3.4 : **Null** balance adjustment of the 741- operational

The calculation of the values of the load resistors $R_{L1,}\ and\ R_{L2}$ as well as the voltage gain $(g_m\ R_c)$ is explained in **Example 3.1.**

Example 3.1

A Differential amplifier that uses NPN – transistors use a ± 15 V –supply. R_e has a value of $1k\Omega$ and resting collector voltage $V_{ce}\ is + 10V$. The Current gain h_{fe} is 50, and the input impedance $h_{ie}\ is\ 900\Omega$. Calculate the values for $R_{L1}\ and\ R_{L2}$ as well as the voltage gain for the amplifier.

Solution :

$$I_e = \frac{V_e}{R_e} = \frac{-15\ V}{1k\Omega} = \ 0{,}015A$$

$$I_C = \frac{I_e}{2} = \frac{0{,}015A}{2} = 0{,}0075\ mA$$

$$R_L = \frac{V_{RI}}{I_C} = \frac{+V_{CC}\ -\ V_{CC}}{I_c} = \frac{+15V - 10V}{0{,}0075A} = 666{,}6\Omega$$

$$Voltage\ gain\ g_m\ R_L = \frac{h_{fe}}{h_{ie}}\ (R_L) = \frac{50}{900}\ (666{,}6\Omega\ ) = 37.03$$

3.5.1 INPUT MODES

The differential amplifier can operate in a single input mode or a differential input mode. In the first case, the signal is applied on only one input while the other is kept at ground potential.

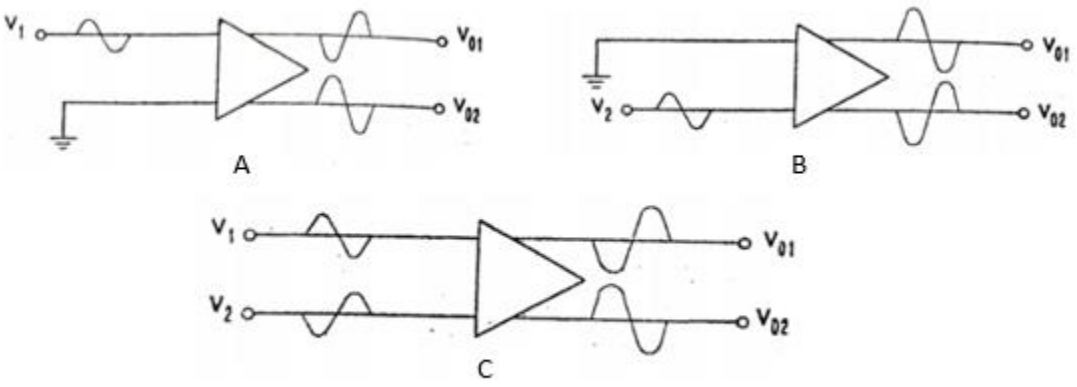


Figure 4.5

This operation is shown in **FIGURE 4.5 (a)** and **(b).** Note the phase relations between the input and output signals. In **figure (c),** the differential input mode is shown.

OPERATION

- We see that V_1 is out of phase and V_2 is in phase with $V_{o1.}$
- An ***operational amplifier is obtained as soon as the difference between*** $\boldsymbol{V_{o1}}$ ***and*** $\boldsymbol{V_{o2}}$ is further amplified V_1 is then the inverting input and V_2 the non $-$ inverting input.

3.6 THE VOLTAGE FOLLOWER

OPERATION

- The ***voltage follower produces a gain of 1*** with no phase inverse.
- ***This circuit reacts more or less than an emitter follower, except*** that the gain is much closer to unity.
- Because the operational amplifier has a high input impedance and a low output impedance, it can be used to match.

- ***It can also be used as a buffer between two stages*** to protect the output of the first stage.

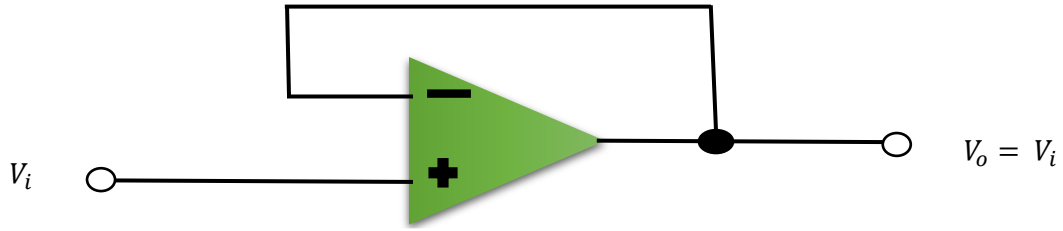
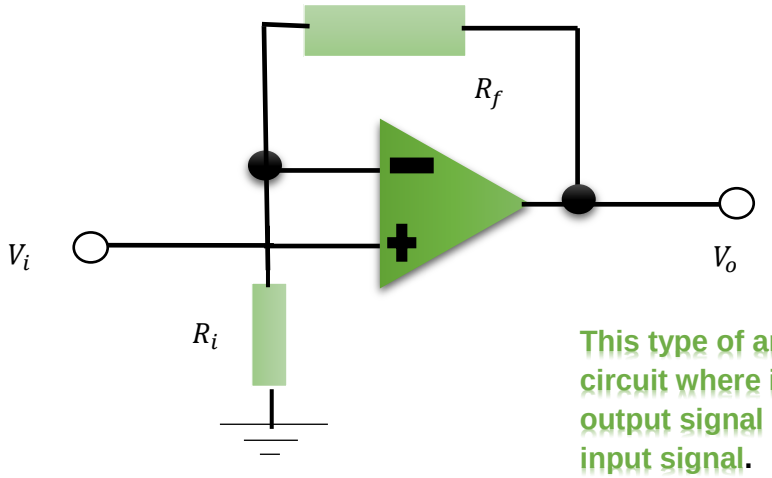


FIGURE 3.8: The voltage follower

3.7 THE NON – INVERTING AMPLIFIER

In the non-inverting amplifier, the input signal is applied on the positive or non-inverting input, which causes the input and output signals to be in phase.

$$V_0 \left(\frac{R_f}{R_i} + 1\right)(V_i)\,, where\ the\ gain\ = \frac{R_f}{R_i} + 1$$



This type of amplifier will be applied in a circuit where it is important that the amplified output signal is in phase with the original input signal.

FIGURE 3.7: The non- inverting amplifier

Example 3.2

Calculate the output if the input voltage $V_i = \pm 10mV$, the input resistor $R_i = 1\ k\Omega$ and the feedback resistor R_f 20 kΩ

Solution:

$$V_o = \left(\frac{R_f}{R_i} + 1\right)(V_i) = \left(\frac{20k\Omega}{1\ k\Omega} + 1\right)(0{,}01V) = 0{,}21V$$

3.8 THE INVERTING AMPLIFIER

The inverting amplifier uses a ***feedback resistor*** R_f which connects the output to the inverse input. It is called negative feedback because the input and output signals are out of ***phase with 180°.*** The relation of ***the feedback resistor*** to the input resistor determines the gain of the amplifier.

Watch the video for more information on amplifiers:
https://www.youtube.com/watch?v=AuZ00cQ0UrE

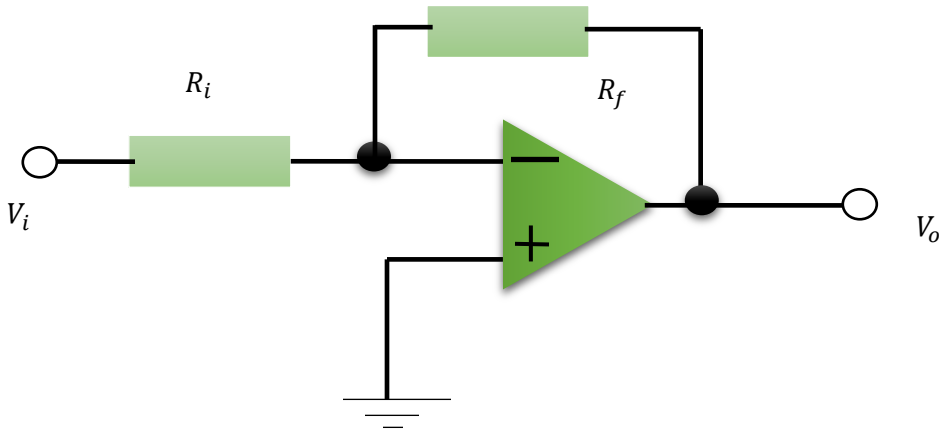


FIGURE 4.6: The inverting amplifier

Example 3.3

Calculate the values of the feedback resistor R_f and the input resistor R if the maximum input and output voltage are ± 2V and ± 12 V, respectively.

Solution:

$$V_o = -\frac{R_f}{R_i}(V_i)$$

$$-\frac{R_f}{R_i} = \frac{V_o}{V_i} = \frac{12}{2} \quad ,therefore\ R_f = 6\ (R_i)$$

If we select a value of 1 kΩ for R_i, R_f must have a value of 12 kΩ
The gain of the amplifier can also be made adjustable if a potentiometer is connected with R_f in series.
If $R_f = R_i$, the minimum gain factor will be 1. An attenuator is obtained as soon as $R_f < R_i$

3.9 THE VOLTAGE ADDER

A voltage adder ***with three Inputs is shown in FIGURE 3.9***. This circuit adds the voltages of which each is multiplied with a constant gain factor. The number of inputs for this circuit can be two or more.

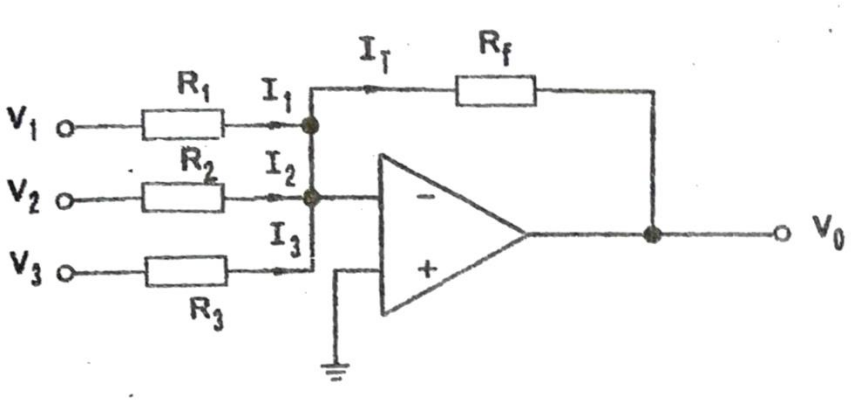


FIGURE 3.9: The voltage adder

The output voltage V_o can be calculated by the following equation :

$$V_o = -\left(\frac{R_f}{R_i}(V_i) + \frac{R_f}{R2}(V_2) + \frac{R_f}{R_3}(V_3)\right)$$

$$Or\ V_o = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)(R_f)$$

$$If\ R_1 = R_2 = R_3 = R_f$$

$$then\ V_o = -(V_1 + V_2 + V_3)$$

This circuit is used in analog computers mostly as well as in servo

3.10 THE INTEGRATOR

FIGURE 3.10 shows an integrator with a capacitor in the feed backline. This circuit produces a slope type output in the form of a triangle wave form. A square form of sine wave form can therefore be transformed to a triangle wave form.

$$V_O(t) = -\frac{1}{RC}\int V_i(t)$$

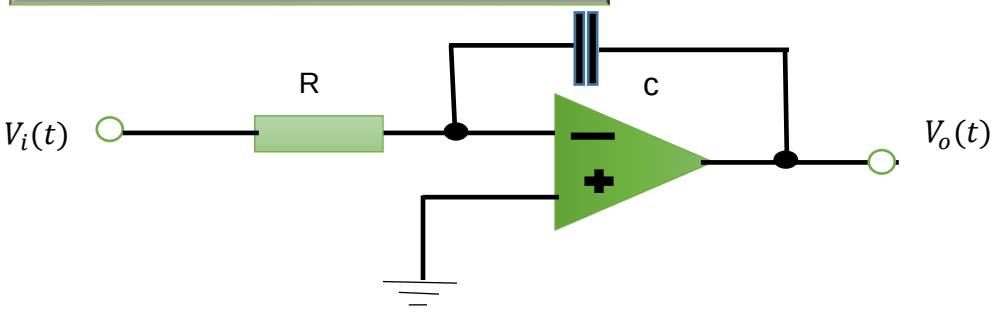


FIGURE 3.10: A basic integrator

If a square wave form is applied to the input, the output will be calculated as followers:

$$\boldsymbol{V_o(t_b) = -\frac{1}{RC}\int_{ta}^{tb} V_i(t_b) + V_{c(t_a)}}$$

Example 3.4

Calculate and draw the output voltage for an integrator with the following values: A square wave form

$V_i = \pm 1V\,, 500Hz$

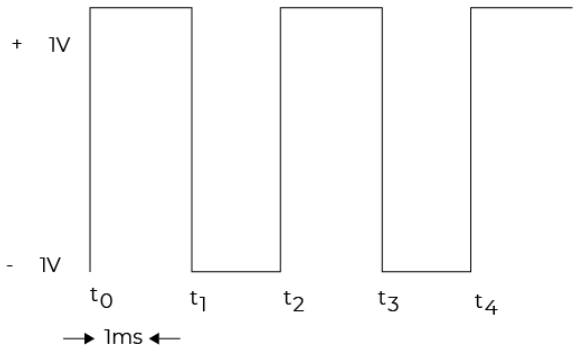
$R = 10M\Omega$

$C = 1\,nF.$

Accept that the capacitor is discharged initially.

Solution:

$$t = \frac{\left(\frac{1}{f}\right)}{2} = \frac{\left(\frac{1}{500}\right)}{2} = 1ms = 0{,}001s$$



$$V_o(t_1) = -\frac{1}{RC}\int_{t0}^{t1} V_i(t_1) \quad + \quad V_{c(t_0)}$$

$$= \frac{1}{10M\Omega \;X\; 1nF}\int_{t0}^{t1} (1)(0{,}001) + 0$$

$$= -\frac{1}{0{,}01}\int_{t0}^{t1} 0{,}001$$

$= -0{,}1\,V\ or - 100mV$

$$V_O(t_2) = -\frac{1}{RC}\int_{t1}^{t2} V_i(t_2) + V_C(t_1)$$

$$= -100\int_{t1}^{t2} (-1)(0{,}001) + (-0{,}1)$$

$$= +0{,}1 - 0{,}1 = 0$$

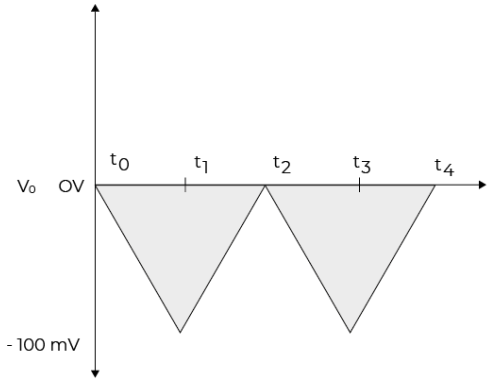


FIGURE 4.12: Output waveform of the integrator

A DC – Deviation originates at low frequencies, which, although it is small, can also be integrated and thus saturates the amplifier. A shunt resistor Rs can be connected in parallel with the capacitor to limit the low-frequency gain of the circuit. An additional resistor R2 can be connected between the non – inverting input and ground to reduce the DC –OFFSET FIGURE 4.13 shows this practical circuit.

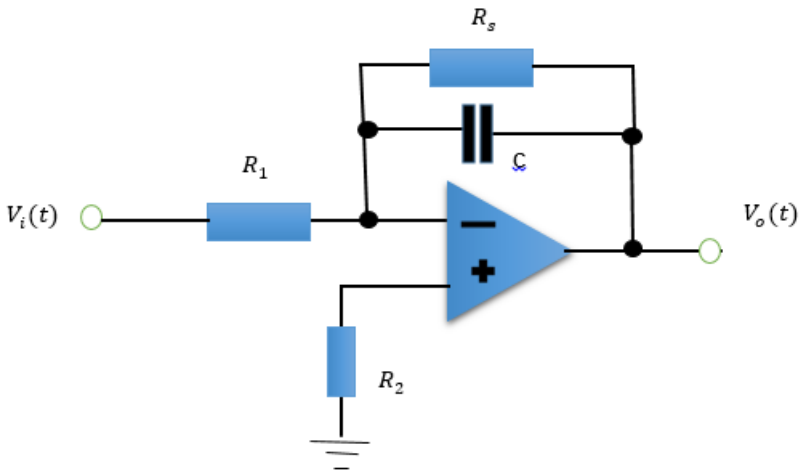


FIGURE 4.13: A practical integrator

$The\ gain\ factor\ of\ the\ circuit\ is\ -\frac{R_s}{R_1}\ and\ the\ value\ R_2\ is\ calculated\ by:\ R_2 = \frac{(R_1)(R_s)}{R_1 + R_s}$

; This circuit however reacts as an integrator for frequencies higher than:

$$f_c = \frac{1}{2\pi R_{sC}}$$

3.11 THE DIFFERENTIATOR

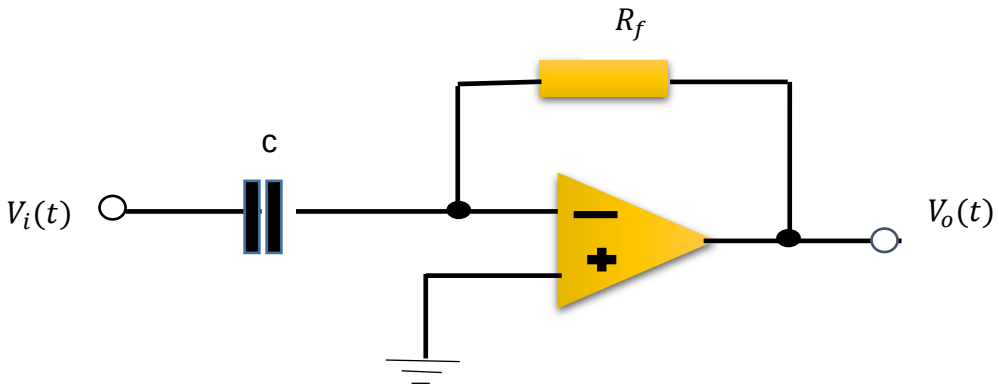


FIGURE 3.14: A basic differentiator

One problem of the basic circuit is that the reactance of the capacitor varies inversely with frequency. Therefore, the amplifier gain increases at high frequencies (gain $\infty R_{f/X_c}$) and makes the circuit acceptable for high-frequency noise. A series resistor relates to the capacitor to limit the high-frequency gain to the factor of $-R/R_f$.

$$V_o(t) = -RC\,\frac{d}{dt}(V_i\,\sin(\omega f))$$

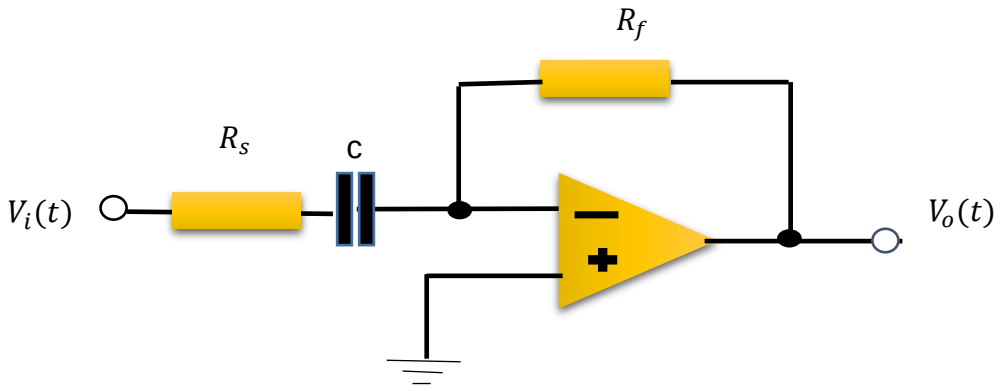


FIGURE 3.15 : A practical

Example 3.5

Calculate the values of $C\ , R_f, and\ R_s$ in a practical differentiator for a 100Hz signal and a high-frequency gain of 10.

Solution :

$$Period\ t = \frac{1}{f} = \frac{1}{100}\ 2 = 0{,}01s$$

$$Theefore\ t = RC\ , 0{,}002 = RC$$

Select a capacitor value of 1 µF : $0{,}01 = R_f\ (1\ X\ 10^{-6})$

$$R_f = \frac{0{,}01}{0{,}000001S} = 100\ 000\Omega$$

$$High\ frequency\ gain = \frac{R_f}{R_s}\ 10\ ;\ \therefore\ R_s = \frac{R_f}{10} = \frac{100\ 000\Omega}{10} = 10k\Omega$$

If the input signal is a sine wave form, the output wave form will experience 90°. A triangle input wave form produces a square wave from on the output.

3.12 THE COMPARATOR

The comparator is a circuit ***that compares an input voltage with a reference voltage***. The output of the comparator indicates ***whether the input signal is above*** or ***below the reference voltage***. If the input voltage of an Inverse Comparator is above the reference voltage, the output approaches the negative supply potential and vice versa. ***A Comparator is used in the open loop mode*** and operates similarly to a differential amplifier. The output is driven close to the supply voltage.

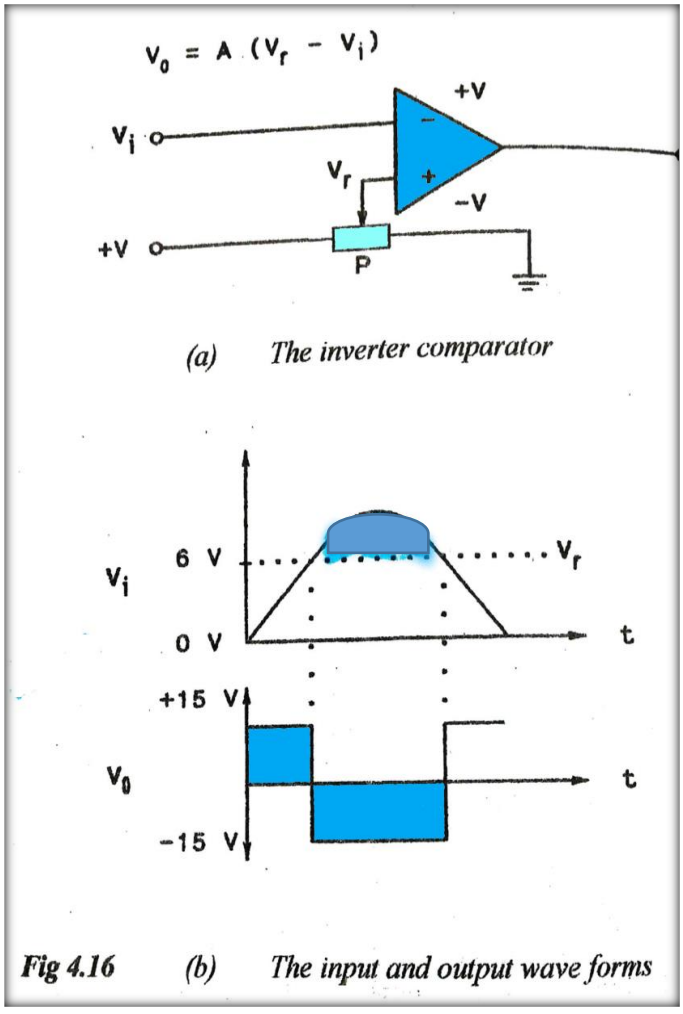


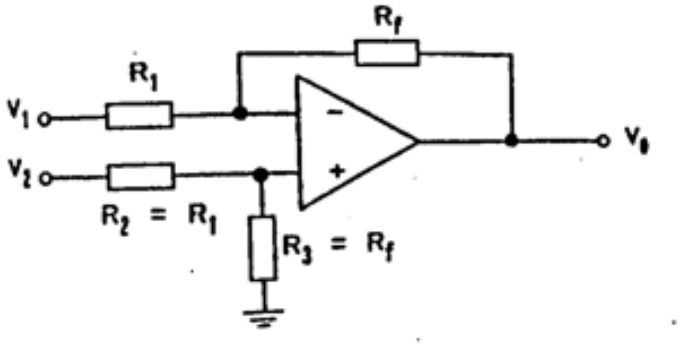
FIGURE 4.16 (b) shows how the comparator reacts with a reference signal of 6 V. Because the output is a square wave form, the circuit can be used as a switch circuit. The potentiometer is used to adjust the reference voltage. If a fixed reference voltage ¡s required, a voltage divider network can be used in place of the potentiometer.

3.13 THE DIFFERENTIAL AMPLIFIER OR SUBTRACTOR

FIGURE 4.17 BELOW shows the differential amplifier, which uses an operational amplifier
This circuit can be used to monitor the difference between two voltage levels.

$$V_o = \frac{R_f}{R_1}(V_2 - V_1)$$

$$If\ R_1 = R_2 = R_3 = R_f\ then\ V_o = V_2 - V_1$$



3.14 ACTIVE FILTERS

A filter is an appliance that passes electrical current at certain frequencies or frequency bands while preventing other frequencies from passing. An active filter is a network that consists of resistors and capacitors around an operational amplifier or transistors.

Active filters generally cost less than passive filters, which use inductors. They can be tuned in or adjusted easily without drastic changes in the design. It also isolates the load from the source because of the impedance differences in the input and output. Lastly, signals are amplified, and therefore no losses are present.

A ***Disadvantage*** of active filters is, other than a passive filter, the supply voltage the circuit requires.

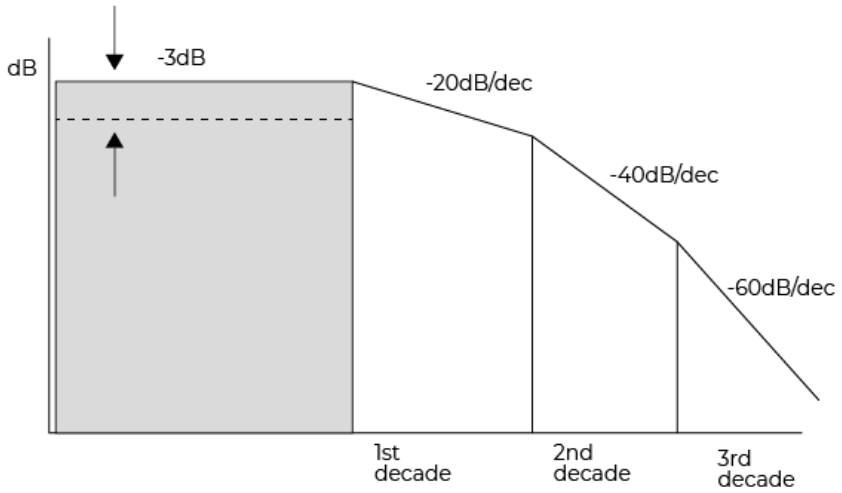
We will look at the four filters: the low – pass , high pass, band pass, and **band reject filters.**

: Watch the video for more information on filters:
https://www.youtube.com/watch?v=gEeF8sEQTEc

Their order also classifies filters, an integrator, which is also called the number of poles. e.g., a two–order filter has two poles. The higher this order, the higher the effectiveness, but the complexity also increases correspondingly.

The frequency response outside the past band of a filter with an order of *n* has a slope with an asymptote of $20n\ dB\ per\ decade\ or\ 6n\ dB\ per\ octave$. This way, the response of a first order$(n = 1)$the filter will drop by 20 dB per decade. See

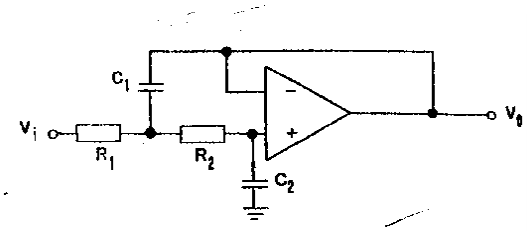
FIGURE 3.19

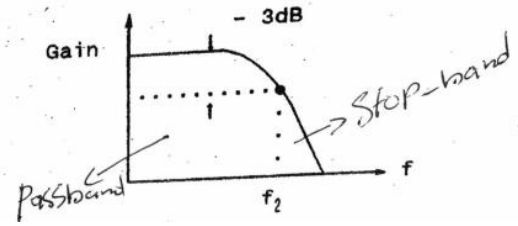


$$If\ R_1 = R_2 = R_3 = R_f\ then\ V_o = V_2 - V_1$$

4.15 THE LOW PASS FILTER

FIGURE 3.20 shows a low – pass filter with a gain of unity and its associated gain response





$$f_o = \frac{1}{2\pi\sqrt{C_1 C_2 \quad R_1 R_2}}$$

FIGURE 3.20

$$If\ C_1 = C_2 = and\ R_1 = R_2,\ then\ f_o = \frac{1}{2\pi CR}$$

3.16 THE HIGH PASS FILTER

FIGURE 3.21 below Shows the "Sallen and key " – high pass filter with a gain of unity and the respective gain response.

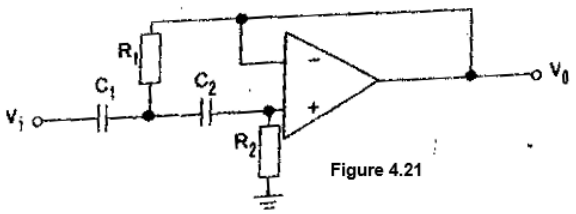
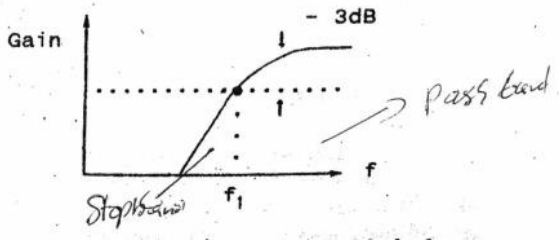


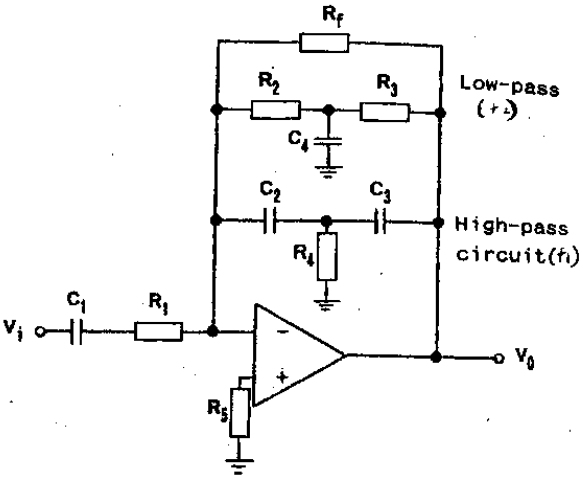
Figure 4.21

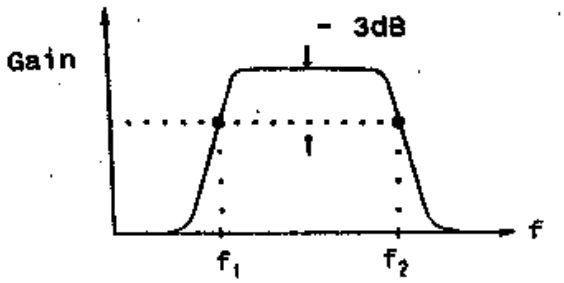


The same equation of the low – pass filter is used to calculate the lower cut – off frequency of the high – pass filter

3.17 THE BAND PASS FILTER

The band – pass filter is shown in **FIGURE 4.22**. It is a more complex circuit which uses the T-type R-C circuits of both the low – and high – pass filters parallel in the feedback line.





The high-pass circuit determines the lower cut-off frequency f_1 while rhe low-**pass** circuit determines the upper cut-off frequency f_1. Refer to the gain response. The values of the filter components are selected as follows:

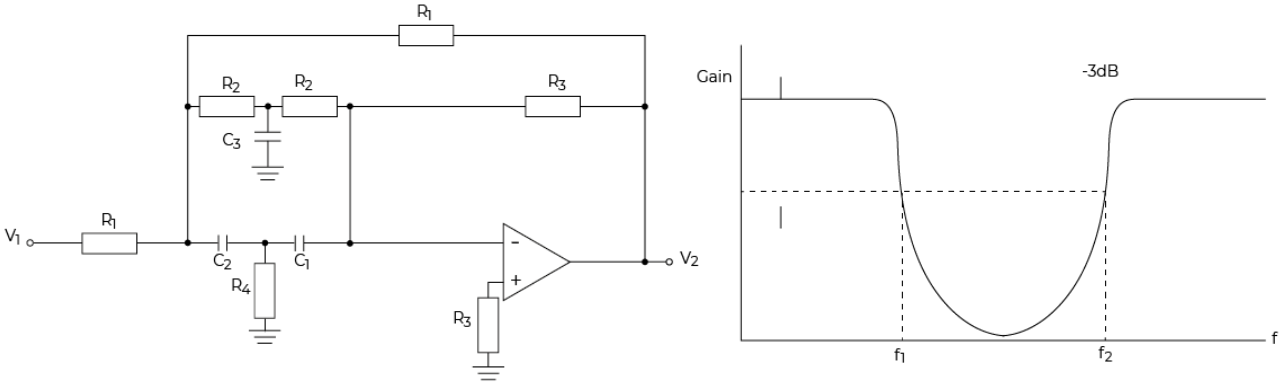
$$\boldsymbol{R = R_2 = R_3 = 2R_4 \; and \; C = C_2 = C_3 = {}^{1}/_{2}\, C_4.} \; \boldsymbol{The \; tuned \; in \; frequency \; is:}$$

$$\boldsymbol{f_0 = \frac{1}{2\pi CR}}$$

3.18 THE BAND – REJECT OR NOTCH FILTER

Below is a circuit diagram of a band-reject filter. The circuit uses two T type circuits connected on the input instead of the feedback line

NOTE: the frequency response in **FIGURE 3.18 (D)** is applicable to this circuit.



$$The \; mid - frequency \; is \; calculated \; by:$$

$$f_O = \frac{1}{2\pi RC} \; ; Where \; R = R_2 = R_3 = 2R_4 \; and \; C = C_1 = C_2 = \frac{1}{2} C_3$$



When working on an operational Amplifier, you need to pay attention to the information that is given i.e., the type of amplifier, the type of the different formulas used for each operational amplifier. This information allows you to be able to apply the correct formula.

SELF CHECK 3

1.) Draw the outlay of a dual-in-line pack of a 741 - operational amplifier and name all the connections.

2.) Mention 5 properties of an ideal operational amplifier.

3.) Draw the simplified circuit diagram of the operational amplifier.

4.) Define the term drift and mention the factors responsible therefor.

5.) Draw the circuit diagram of a differential amplifier that uses NPN-transistors and briefly describe the operation of the circuit.

6.) Briefly describe, with the aid of a circuit diagram, how you can eliminate the differential input offset in an operational amplifier with the aid of a potentiometer

7.) Calculate the values of $\mathrm{R_f}$ and $\mathrm{R_S}$ in a practical differentiator, if a 1 kHz signal is applied and a high-frequency gain of 10 is required, Choose a capacitor of $1\mu\mathrm{F}$.

8.) The gain factor of a practical integrator in FIGURE 4.13 is 10, and the input resistance is 1kΩ. Calculate the value of $\mathrm{R_2}$ as well as the frequency from where the circuit will react as an integrator. Assume that C = 10 nF.

9.) Show the input and output wave forms of an operational comparator of which the analog input varies between 0V and 10 V and the reference input is set to 4 V

10.) A differential amplifier that contains NPN transistors is supplied with a ± 12 dual-voltage Supply: $R_C\ 2k\Omega\ , V_{cc} = +8V\ and\ h_{ie}\ 900\Omega$.

11.1.) Calculate the values of $R_{L1}\ and\ R_{L2}$

11.2.) Calculate The current gain h_fo that the transistors should have to have a voltage gain of 10.

11.) Draw the circuit diagrams of the following active filters:

- Low-pass filter
- High-pass filter
- Band-pass filter
- Band-reject filter

.

12.) Calculate the resistor values for a high-pass filter if the mid-frequency should be 100 kHz, and Capacitor values of 10 nF are chosen.[2]

References

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Available from : https://www.youtube.com/watch?v=P33-CQlwKLQ[accessed 20th March 2021]